
URS: A Unified Neural Routing Solver for Cross-Problem Zero-Shot Generalization

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Abstract

Multi-task neural routing solvers have emerged as a promising paradigm for their ability to solve multiple vehicle routing problems (VRPs) using a single model. However, existing neural solvers typically rely on predefined problem constraints or require per-problem fine-tuning, which substantially limits their zero-shot generalization ability to unseen VRP variants. To address this critical bottleneck, we propose URS, a unified neural routing solver that achieves zero-shot generalization across a wide range of unseen VRPs with a single model. We propose a unified data representation (UDR) that replaces problem enumeration with data unification, thereby broadening the problem coverage and reducing reliance on domain expertise. In addition, we introduce a mixed bias module (MBM) during encoding to improve node embeddings, which efficiently captures multiple priors inherent to various problems. On top of the UDR, we develop a problem-conditioned parameter generator to further improve zero-shot generalization. Extensive experiments show that URS consistently produces high-quality solutions for 110 VRP variants (including 99 unseen variants) while demonstrating impressive scalability to large-scale instances with up to 7000 nodes. To the best of our knowledge, URS is the first neural solver to handle over 100 VRP variants with a single model. Our code is available at <https://github.com/CIAM-Group/URS>.

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1. Introduction

The Vehicle Routing Problem (VRP) is an essential class of combinatorial optimization problems (COPs) with extensive applications in logistics and supply chain management (Tiwari & Sharma, 2023; Sar & Ghadimi, 2023). Solving VRPs efficiently is challenging due to their NP-hard nature. While exact solvers can produce optimal solutions, they often become computationally prohibitive for real-world scenarios. In recent decades, classical heuristic solvers (Helsgaun, 2017; Vidal, 2022) have achieved impressive performance within acceptable timeframes. Nevertheless, these methods require considerable domain expertise to design specialist rules for each routing problem. Given the growing diversity of VRP variants in real-world applications, manually crafting tailored rules for every case has become impractical.

In recent years, neural combinatorial optimization (NCO) methods have attracted substantial attention for their potential to reduce reliance on handcrafted rules while maintaining competitive solution quality (Bengio et al., 2021; Ba et al., 2026). This has led to the development of many high-performing neural routing solvers that automatically learn implicit problem-specific rules from data under different training paradigms such as supervised learning (SL) (Drakulic et al., 2023; Luo et al., 2023; 2025b), reinforcement learning (RL) (Kool et al., 2019; Zhou et al., 2026a), and self-improved learning (SIL) (Luo et al., 2025a; Pirnay & Grimm, 2024). Although these solvers have demonstrated impressive performance on specific problems (Huang et al., 2025; Zhou et al., 2026b), they typically require architectural customization and per-problem retraining to accommodate the distinct constraints and features of different VRP variants, thereby increasing overall training costs and hindering practical deployment to new problems.

To address the challenge of cross-problem generalization, growing attention has been directed toward multi-task learning capable of handling diverse routing problems. As summarized in Table 1, existing efforts largely fall into two categories: 1) constraint combination-based multi-task learning and 2) adapter-based fine-tuning. In the first approach, VRP variants are treated as different combinations of constraint attributes, and a unified model is trained across these combinations to enable knowledge sharing for problems with

Table 1. Comparison between our URS and existing neural solvers with multi-task learning. Note that “#VRP Variants” and “Generalizable Scale” refer to the total number of tested problems and the maximum scale reported in the original papers, respectively.

Multi-task Neural Routing Solver	Training Paradigm	#VRP Variants	Symmetric VRPs	Asymmetric VRPs	Pickup and Delivery Problems	Zero-shot Generalization	Training Scale	Generalizable Scale	Remarks
MTPOMO (Liu et al., 2024)	RL	16	✓	×	×	✓	100	1000	Constraint Combination
MVMoe (Zhou et al., 2024)	RL	16	✓	×	×	✓	100	1000	Constraint Combination
RouteFinder (Berto et al., 2025)	RL	48	✓	×	×	✓	100	1000	Constraint Combination
CaDA (Li et al., 2025a)	RL	16	✓	×	×	✓	100	200	Constraint Combination
MTL-KD (Zheng et al., 2025)	RL+KD	16	✓	×	×	✓	100	1000	Constraint Combination
TSP-FT (Lin et al., 2024)	RL	4	✓	×	×	×	100	1000	Adapter-based Fine-tuning
MTL-MAB (Wang et al., 2025a)	RL	3	✓	×	×	×	100	1000	Adapter-based Fine-tuning
GOAL (Drakulic et al., 2025)	SL	10	✓	✓	×	×	100	1000	Adapter-based Fine-tuning
URS (Ours)	RL	110	✓	✓	✓	✓	100	7000	Unified Data Representation

seen constraints (Liu et al., 2024; Zhou et al., 2024; Li et al., 2025a; Berto et al., 2025; Zheng et al., 2025). The second approach builds a shared model backbone for all problems and incorporates problem-specific input/output adapters, thereby reducing retraining costs (Lin et al., 2024; Drakulic et al., 2025; Wang et al., 2025a).

Despite the above advancements, existing methods still fall short in cross-problem zero-shot generalization. The problem coverage of constraint combination-based approaches is inherently bounded by their manually specified constraint sets, whereas adapter-based approaches require additional fine-tuning and thus cannot perform zero-shot generalization. Furthermore, both strategies fundamentally rely on the explicit problem enumeration through a predefined set of problem tags. This practice is challenging because the constraint space of VRP variants is open-ended and compositional. More critically, creating and maintaining such a problem taxonomy requires considerable domain expertise, which NCO always aims to avoid. A detailed review of related work on single-task and multi-task learning is provided in Appendix A.

In this paper, we propose a powerful **Unified Neural Routing Solver (URS)** to improve the cross-problem zero-shot generalization ability for NCO methods. Our contributions can be summarized as follows: (1) We propose a unified data representation (UDR) that replaces existing problem enumeration with data unification, significantly broadening problem coverage while mitigating reliance on domain-specific expertise; (2) We introduce a mixed bias module (MBM) during encoding to improve node embeddings, which efficiently captures multiple priors inherent to various problems while reducing architectural redundancy; (3) On top of the UDR, we develop a problem-conditioned parameter generator to further improve zero-shot generalization; (4) Extensive experiments on **110** VRP variants show that URS not only achieves competitive performance against specialist neural solvers on seen variants but also exhibits strong zero-shot generalization across **99** unseen variants. Notably, URS demonstrates impressive scalability to large-scale instances with up to 7000 nodes. To the best of our knowledge, URS is the first neural solver to efficiently solve over 100 VRP variants with a single model, without retraining or fine-tuning.

2. Preliminaries

In this section, we first introduce the definition of VRP, then provide an overview of recent constructive neural solvers for solution generation (Kool et al., 2019; Kwon et al., 2020).

Vehicle Routing Problems Consider a VRP instance with an optional depot indexed by 0 and n customers indexed by $\{1, 2, \dots, n\}$, such as in the Capacitated VRP (CVRP). The instance can be represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the node set $\mathcal{V} = \{v_i\}_{i=0}^n$ has a total size of $|\mathcal{V}| = 1 + n$ unless the variant has no depot (e.g., the Traveling Salesman Problem), and the edge set is $\mathcal{E} = \{e(v_i, v_j) \mid v_i, v_j \in \mathcal{V}, v_i \neq v_j\}$. The travel cost between nodes is given by a distance matrix $\mathbf{D} = \{d_{ij} \mid \forall i, j \in 0, \dots, n\}$. In VRP, each node $v_i \in \mathcal{V}$ includes node coordinates $\{x_i, y_i\}$ when available and problem-specific attributes (e.g., demands in CVRP). Formally, a feasible solution is denoted by a node permutation $\pi = (\pi_1, \pi_2, \dots, \pi_m)$ (Toth & Vigo, 2002) (i.e., a finite sequence) that represents a set of valid routes. Each route is traversed by a single vehicle originating and terminating at its own depot or starting node, while strictly satisfying all constraints. Let Ω denote the feasible sequence set, given an objective function $f(\pi|\mathcal{G})$, we aim to search for a sequence π^* with the maximum objective, i.e., $\pi^* = \arg \max_{\pi \in \Omega} f(\pi|\mathcal{G})$. In most VRPs, $f(\pi|\mathcal{G})$ can be defined as the negative value of the total distance of π .

In this paper, we evaluate URS across 110 VRP variants spanning 10 distinct tags, in which each variant may involve either a single tag or a combination of the following tags: (1) Capacity (C); (2) Open Route (O); (3) Backhaul (B); (4) Backhaul and Priority (BP); (5) Duration Limit (L); (6) Time Windows (TW); (7) Multi-Depot (MD); (8) Prize Collecting (PC); (9) Asymmetric (A); and (10) Pickup and Delivery (PD). Detailed definitions are provided in Appendix B.

Constructive Neural Routing Solver Most constructive NCO methods employ an encoder-decoder architecture for solution construction (Luo et al., 2023; Kwon et al., 2020). Let learnable model parameters be denoted as $\theta = \{\theta_{enc}, \theta_{dec}\}$. Without loss of generality, we present the prevailing autoregressive construction pipeline using AM (Kool et al., 2019). Given an instance $\mathcal{G}$, raw node

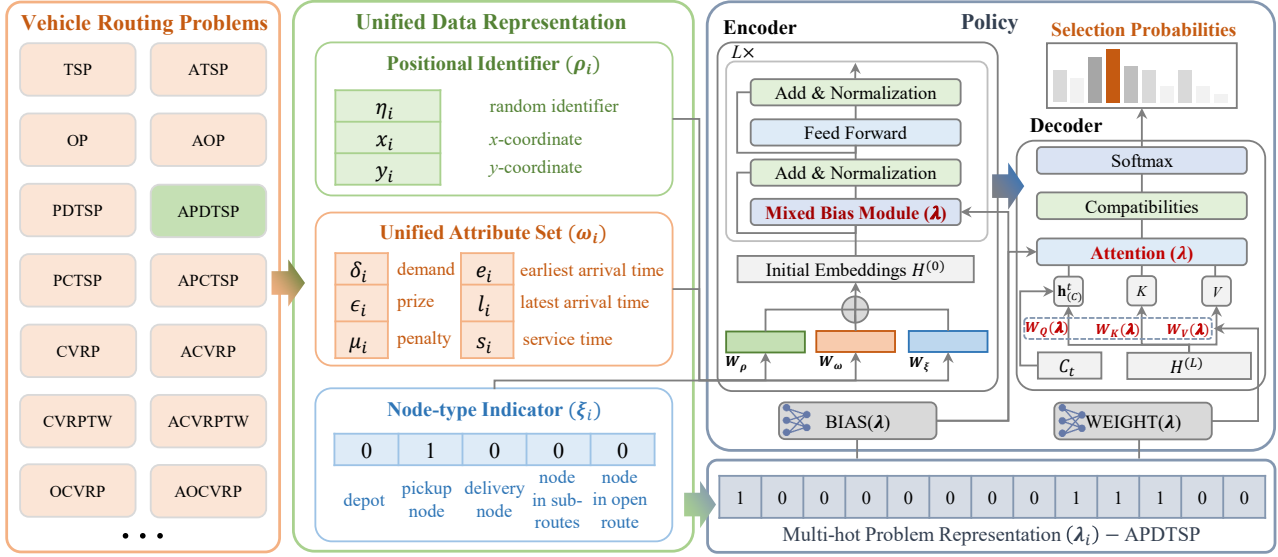


Figure 1. The pipeline of our URS for solving 110 VRP variants using a single model without any fine-tuning, illustrated on APDTPSP.

features are first mapped by a linear projection to initial embeddings $H^{(0)} = \{\mathbf{h}_i^{(0)}\}_{i=0}^n$. They are then processed by an encoder θ_{enc} with L attention layers to produce refined node embeddings $H^{(L)} = \{\mathbf{h}_i^{(L)}\}_{i=0}^n$. At each decoding step t , the decoder θ_{dec} sequentially selects a node to append to the current partial solution $\pi_{1:t-1} = (\pi_1, \pi_2, \dots, \pi_{t-1})$, where $\pi_1, \pi_{t-1} \in \mathcal{V}$ are the **first** and **last** visited node, respectively. This sequential process continues until a complete solution is formed. To guarantee the feasibility of generated solutions, a masking function $\mathcal{M}_t$ sets the selection probabilities of the following nodes to $-\infty$ during the construction process: nodes that (1) are visited or (2) violate problem-specific constraints (e.g., capacity).

3. Methodology

In this section, as illustrated in Figure 1, we propose a powerful Unified Neural Routing Solver (URS), which significantly improves cross-problem zero-shot generalization for NCO methods. Its key components are elaborated below.

3.1. Unified Data Representation

Existing multi-task neural solvers (Zhou et al., 2024; Drakulic et al., 2025) fundamentally rely on explicit problem enumeration via a predefined set of problem tags, which is challenging because the constraint space of VRPs is open-ended and compositional. More critically, creating and maintaining such a problem taxonomy requires considerable domain expertise, which NCO always aims to avoid.

From a data perspective, despite the substantial diversity in constraints, their instances share a common structural representation. For existing constraint combination-based meth-

ods (Liu et al., 2024; Zhou et al., 2024), the problems they evaluate can be formulated as CVRPTW instances. Leveraging this structural commonality, we propose a unified data representation (UDR) $\mathcal{U} = \{\mathbf{u}_i\}_{i=0}^n$. By replacing explicit problem enumeration with data unification, our UDR broadens problem coverage and reduces reliance on domain expertise. Notably, the inputs to existing multi-task neural solvers based on either problem-specific adapters (Drakulic et al., 2025) or constraint combinations (Zhou et al., 2024) can be viewed as special subsets of our proposed UDR.

Instead of relying on a predefined problem taxonomy with a discrete set of problem tags, UDR decouples instance representation from constraint definitions and operates in a unified space characterized by both continuous and discrete features. It allows a single neural solver to address a much larger, open-ended space of VRP variants, as new problems can be seamlessly incorporated into this unified representation. Problem-specific constraints are delegated to the model-agnostic masking function $\mathcal{M}_t$. For example, for all CVRP variants, only the remaining load is explicitly provided as input, while diverse additional constraints (e.g., time windows) are enforced implicitly by masking infeasible candidate nodes during solution construction, rather than being fully enumerated in the decoder’s input. Thus, UDR enables a single model to handle various VRP variants while significantly reducing reliance on domain expertise. Specifically, for an instance $\mathcal{G}$, each $\mathbf{u}_i = \{\rho_i, \omega_i, \xi_i\}$, where ρ_i, ω_i, ξ_i are a positional identifier, unified attribute set, and node-type indicator, respectively.

Positional Identifier (ρ_i) We define a unified positional identifier for each node as $\rho_i = \{\eta_i, x_i, y_i\} \in [0, 1]^3$. The $\eta_i \sim \text{Uniform}(0, 1) \in \mathbb{R}^1$ is a sampled scalar, which is

used to address asymmetric problems, following Drakulic et al. (2023). For symmetric cases, we simply set $\eta_i = 0$.

Unified Attribute Set (ω_i) We define the unified attribute set as $\omega_i = \{\delta_i, \epsilon_i, \mu_i, e_i, l_i, s_i\}$, where they correspond to demand, prize, penalty, earliest arrival time, latest arrival time, and service time, respectively. The unification empowers the model to learn the relative importance of individual attributes in a shared embedding space. It also supports effortless attribute extension or ablation via simple zero-filling, eliminating the need to change the architecture.

Node-type Indicator (ξ_i) We provide a 5-dimensional binary vector $\xi_i \in \{0, 1\}^5$ for each node, encoding: depot, pickup node, delivery node, node in sub-routes, node in open route. The nodes in sub-routes and open routes mean the solution π can have sub-routes or be an open route. We introduce explicit structural roles to enable the model to generalize over diverse variants. Further ablation analysis concerning ξ_i is provided in Appendix K.1.

Multi-hot Problem Representation (λ_i) Through the UDR, each problem instance activates only a specific subset of the unified feature space. For any instance $\mathcal{G}_i$ drawn from different problems, a corresponding multi-hot problem representation λ_i is obtained by indicating active (non-zero) feature slots with a value of 1. This representation captures only the presence of features, in contrast to their raw values, and it subsequently provides conditional guidance to the position bias weight (see Equation (6)) and the adaptive decoder parameters $\theta_{dec}(\lambda)$ (see Equation (9) and Equation (10)), thereby helping the model in distinguishing problem variants and refining its node selection process.

A more detailed description of U and λ is provided in Appendix C and Appendix D, respectively.

3.2. Model Architecture

As shown in Figure 1, we adopt AM (Kool et al., 2019) as our basic model. While UDR provides cross-problem consistency, different problems have diverse geometric priors. To efficiently learn the multiple priors inherent in various problems and obtain better cross-problem zero-shot generalization, we propose two key enhancements: (1) a mixed bias module (MBM) that captures multiple priors inherent to various problems, and (2) problem-conditioned parameter generators conditioned on the UDR. Their implementations are detailed below.

Embedding Layer Given an instance $\mathcal{G}$, for each $\mathbf{u}_i = \{\rho_i, \omega_i, \xi_i\}$, we project it into d -dimensional embeddings through three separate linear transformations:

$$\mathbf{h}_i^{(0)} = \rho_i W_\rho + \omega_i W_\omega + \xi_i W_\xi, \quad (1)$$

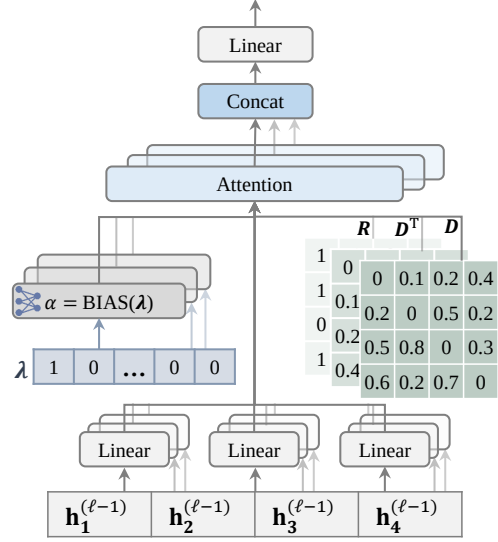


Figure 2. The structure of the proposed mixed bias module, explained on a 4-node APDTSP instance. We replace plain attention with MBM during encoding to improve node embeddings, which effectively captures multiple priors inherent to various problems while reducing architectural redundancy.

where $W_\rho \in \mathbb{R}^{3 \times d}$, $W_\omega \in \mathbb{R}^{6 \times d}$, $W_\xi \in \mathbb{R}^{5 \times d}$ are learnable matrices. Then we obtain a set of initial embeddings $H^{(0)} = \{\mathbf{h}_i^{(0)}\}_{i=0}^n$ for all nodes in instance $\mathcal{G}$. This embedding is the initial input of encoder θ_{enc} . After passing through L stacked attention layers, $H^{(0)}$ is transformed into advanced node embeddings $H^{(L)} = \{\mathbf{h}_i^{(L)}\}_{i=0}^n$. The detailed encoding process is provided in Appendix E.1.

Mixed Bias Module Given the distinct priors across VRP variants, efficiently capturing the intrinsic geometric bias is pivotal for enhancing the cross-problem generalization of RL-based multi-task neural solvers. Current leading RL-based models generally adopt a heavy encoder and light decoder architecture, where the quality of the encoder-generated node embeddings plays a key role in overall performance. While prior studies like MatNet (Kwon et al., 2021) have attempted to handle asymmetry by integrating directional distance biases via parallel layers, such designs incur high model complexity and lack universality. Consequently, effectively encoding multiple inherent biases, such as symmetric and asymmetric distances, and relational dependencies, remains challenging, hindering the acquisition of high-quality solutions across a wide range of problems.

To address the limitation, we propose the mixed bias module (MBM) to replace the vanilla attention mechanism (Vaswani et al., 2017), which enhances cross-problem generalization while reducing architectural redundancy. As shown in Figure 2, MBM can efficiently capture three problem-specific prior matrices via three separate attention calculations: (1) an outgoing distance matrix D ; (2) an incoming

distance matrix D^T ; and (3) an optional relation matrix $R = \{r_{ij} | \forall i, j \in 0, \dots, n\}$, where $r_{ij} = 0$ if a predefined relation exists (e.g., pickup-delivery pairing) and 1 otherwise. This ensures that smaller values are preferred, aligning with the distance metric, where smaller values correspond to a larger selection bias. For the ℓ -th layer, the MBM output $\hat{\mathbf{h}}_i^{(\ell)}$ can be expressed as:

$$\bar{\mathbf{h}}_i^{(0)} = \text{Attention}(\mathbf{h}_i^{(\ell-1)}, H^{(\ell-1)}, f(\alpha, N, D_i)), \quad (2)$$

$$\bar{\mathbf{h}}_i^{(1)} = \text{Attention}(\mathbf{h}_i^{(\ell-1)}, H^{(\ell-1)}, f(\alpha, N, D_i^T)), \quad (3)$$

$$\bar{\mathbf{h}}_i^{(2)} = \begin{cases} \text{Attention}(\mathbf{h}_i^{(\ell-1)}, H^{(\ell-1)}, f(\alpha, R_i)) & \text{if } R \neq \emptyset \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad (4)$$

$$\hat{\mathbf{h}}_i^{(\ell)} = [\bar{\mathbf{h}}_i^{(0)}, \bar{\mathbf{h}}_i^{(1)}, \bar{\mathbf{h}}_i^{(2)}] W^O, \quad (5)$$

where $[\cdot, \cdot]$ denotes the horizontal concatenation operator, $W^O \in \mathbb{R}^{(3 \times d) \times d}$ is a learnable matrix. For $\text{Attention}(\cdot)$, we adopt the attention mechanism introduced by Zhou et al. (2026a), which boosts the perception of diverse geometric patterns through an adaptation function $f(\alpha, N, d_{ij})$ with bias weight α (see Appendix F for implementation details). Note that since the relation r_{ij} is scale-independent, we remove the scale N in the adaptation function for R . If R is absent, we substitute a zero vector for $\bar{\mathbf{h}}_i^{(2)}$. For a comparison between our MBM and existing attentions (Vaswani et al., 2017; Kwon et al., 2021), please see Appendix G.

Problem-Conditioned Parameter Generation Using a single static set of parameters to solve 110 VRP variants is a significant challenge. To mitigate this, we introduce a problem-conditioned mechanism based on problem representation λ , which consists of two components: (1) $\text{BIAS}(\lambda)$ adjusts the degree to which different priors are integrated into the attention; and (2) $\text{WEIGHT}(\lambda)$ generates decoder parameters for each problem.

As shown in Figure 3, unlike existing adapter-based fine-tuning methods (Lin et al., 2024; Drakulic et al., 2025), we provide a unified decoding entrance and enable URS to adaptively generate decoder parameters for each problem. This conditional weight generation removes problem-specific adapters, enabling zero-shot generalization to unseen problems while maintaining solution quality.

For MBM in Equation (5), we generate α via a lightweight bias network $\text{BIAS}(\lambda)$:

$$\text{BIAS}(\lambda) = \max(1, (\lambda W_1 + \mathbf{b}_1) W_2 + \mathbf{b}_2), \quad (6)$$

where $W_1 \in \mathbb{R}^{|\lambda| \times d}$, $W_2 \in \mathbb{R}^{d \times 1}$, $\mathbf{b}_1 \in \mathbb{R}^d$, $\mathbf{b}_2 \in \mathbb{R}^1$ are learnable parameters. We set the minimum bias to 1, resulting in faster model convergence.

For decoder $\theta_{dec}(\lambda)$, all parameters are generated based on λ . Following Kwon et al. (2020), given the first and last

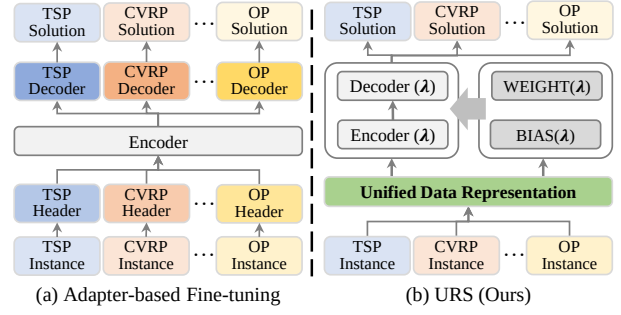


Figure 3. Comparison between our URS and existing adapter-based fine-tuning frameworks.

visited node embeddings $\mathbf{h}_{\pi_1}^{(L)}$ and $\mathbf{h}_{\pi_{t-1}}^{(L)}$, and an optional constraint state $C_t \in \mathbb{R}^1$ (e.g., remaining load), the context embedding $\mathbf{h}_{(C)}^t$ can be expressed as:

$$\mathbf{h}_{(C)}^t = [\mathbf{h}_{\pi_1}^{(L)}, \mathbf{h}_{\pi_{t-1}}^{(L)}, C_t] W_Q(\lambda), \quad (7)$$

where $[\cdot, \cdot]$ is the horizontal concatenation operator, $W_Q(\lambda) \in \mathbb{R}^{(2d+1) \times d}$ is a linear projection matrix conditioned on λ . Missing C_t is zero-padded (see Appendix E.3 for details). The new context embedding $\hat{\mathbf{h}}_{(C)}^t$ is obtained via $\text{Attention}(\cdot)$ (Zhou et al., 2026a) on $\mathbf{h}_{(C)}^t$ and $H^{(L)}$:

$$K = H^{(L)} W_K(\lambda), \quad V = H^{(L)} W_V(\lambda), \quad (8)$$

$$\hat{\mathbf{h}}_{(C)}^t = \text{Attention}(\mathbf{h}_{(C)}^t, K, V, \mathcal{M}_t, f(\alpha, N, d_{t,i})), \quad (9)$$

where $W_K(\lambda), W_V(\lambda) \in \mathbb{R}^{d \times d}$ are linear projection matrices conditioned on λ . Finally, we compute probabilities for feasible nodes $p_\theta(\pi_t = i | \mathcal{G}, \pi_{1:t-1}) = \text{softmax}(\mathbf{u})$ based on the improved compatibility (Zhou et al., 2026a):

$$u_i^t = \begin{cases} \zeta \cdot \tanh\left(\frac{\hat{\mathbf{h}}_{(C)}^t (\mathbf{h}_i^{(L)})^T}{\sqrt{d}}\right) + f(\alpha, N, d_{t,i}) & \text{if } i \notin \{\pi_{1:t-1}\} \\ -\infty & \text{otherwise,} \end{cases} \quad (10)$$

where ζ is the clipping parameter. Inspired by Lin et al. (2022), we use a simple MLP hypernetwork $\text{WEIGHT}(\lambda)$ to generate $\{W_Q(\lambda), W_K(\lambda), W_V(\lambda)\}$ (see Appendix E.2 for implementation details). For α in Equation (9) and Equation (10), we generate them via $\text{BIAS}(\lambda)$ in Equation (6).

4. Experiments

In this section, we present a comprehensive evaluation of URS against both classical and neural solvers. For URS, we focus on four key aspects: (1) performance on 11 seen problems; (2) zero-shot generalization on 99 unseen problems; (3) scalability to large-scale VRP variants; and (4) results on benchmark datasets. All experiments are conducted using a single NVIDIA GeForce RTX 4090 GPU (24GB of memory) for neural solvers and an Intel Xeon Gold 6240R CPU (2.40 GHz) for classical solvers.

Table 2. Performance comparison on 8 seen less-explored variants.

Method	OP100		PCTSP100		PDTSP100		ACVRP100		CVRPTW100		CVRPB100		OCVRP100		OCVRPTW100	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	1.5m	0.00%	1.2h	0.00%	9.8m	0.00%	6.6m	0.00%	19.6m	0.00%	20.8m	0.00%	5.3m	0.00%	20.8m
Sym-NCO	0.68%	15s	0.14%	16s	—	—	—	—	—	—	—	—	—	—	—	—
BQ	0.63%	7s	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Heter-AM	—	—	—	—	6.61%	5m	—	—	—	—	—	—	—	—	—	—
MVMoE	—	—	—	—	—	—	—	—	4.90%	15s	1.28%	11s	3.14%	13s	3.85%	15s
MTPOMO	—	—	—	—	—	—	—	—	5.31%	8s	1.67%	5s	3.46%	6s	4.41%	7s
ReLD-MTL	—	—	—	—	—	—	—	—	4.56%	10s	0.90%	6s	2.32%	7s	3.10%	9s
GOAL-MTL	1.20%	38s	—	—	—	—	—	—	4.66%	42s	—	—	—	—	—	—
URS-STL	0.11%	4s	-0.09%	5s	3.22%	4s	2.01%	1.4m	4.50%	8s	1.59%	6s	2.25%	6s	3.22%	8s
URS	0.45%	4s	1.06%	5s	4.98%	4s	3.06%	1.4m	6.13%	8s	1.46%	6s	3.24%	6s	5.07%	8s

Table 3. Experimental results on seen ATSP, TSP, and CVRP.

Method	ATSP100		TSP100		CVRP100	
	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	2m	0.00%	6m	0.00%	9.1m
Sym-NCO	—	—	0.14%	5s	1.46%	6s
BQ	1.27%	1m	0.35%	7s	3.24%	8s
LEHD	—	—	0.59%	2s	3.68%	2s
POMO	—	—	0.13%	5s	1.25%	6s
ICAM	3.05%	1m	0.15%	4s	2.04%	4s
MatNet	0.95%	6.5m	—	—	—	—
MVMoE	—	—	—	—	1.65%	12s
MTPOMO	—	—	—	—	1.85%	6s
ReLD-MTL	—	—	—	—	1.42%	8s
GOAL-MTL	1.77%	1m	—	—	4.22%	48s
URS-STL	0.62%	1.1m	0.08%	6s	1.63%	6s
URS	2.26%	1.1m	0.57%	6s	1.81%	6s

4.1. Experimental Setup

Problem Setting We train URS on 11 mixed variants to ensure that most features in UDR have been seen at least twice, which prevents URS from overfitting a specific feature to a single problem. Training problems and related data generations are as follows: (1) ATSP (Kwon et al., 2021); (2) TSP (Kool et al., 2019); (3) CVRP (Kool et al., 2019); (4) ACVRP (Kwon et al., 2021; Kool et al., 2019); (5) Orienteering Problem (OP) (Kool et al., 2019); (6) PCTSP (Kool et al., 2019); (7) PDTSP (Li et al., 2021); (8) CVRPTW (Zhou et al., 2024); (9) OCVRP (Zhou et al., 2024); (10) CVRPB (Zhou et al., 2024), and (11) OCVRPTW (Zhou et al., 2024). We provide further discussion of problem selection in Appendix J.1. Each of 110 variants comprises 1,000 instances. The related oracle solvers and problem representations are detailed in Appendix D.

Model & Training Setting We use an embedding dimension d of 128 and a feed-forward layer dimension of 512. We set the number of attention layers in the encoder to 12. The clipping parameter ζ is set to 50 in Equation (10). URS is trained by the REINFORCE (Williams, 1992) algorithm, and the data is generated on the fly (see Appendix E.4 for

more details). We train URS for 500 epochs with 2,000 batches per epoch, and each batch has a size of 128. At each training step, we randomly select one problem from 11 problems. We employ the AdamW optimizer (Loshchilov & Hutter, 2017) with an initial learning rate of 10^{-4} , which is decayed by a factor of 0.1 at the 451st epoch. Weight decay is set to 10^{-6} . More details are provided in Appendix H.

Baseline We compare URS with the following methods: (1) **Classical Solvers:** PyVRP (Wouda et al., 2024), LKH3 (Helsgaun, 2017), and OR-Tools (Perron & Furnon, 2023). We run LKH3 (Helsgaun, 2017) with 10000 trials and 1 run (Kool et al., 2019). For PyVRP and OR-Tools, we run them on a single CPU core with time limits of 20s (Li et al., 2025a); (2) **Neural Solvers:** (i) *specialist solvers:* Sym-NCO (Kim et al., 2022), BQ (Drakulic et al., 2023), LEHD (Luo et al., 2023), ICAM (Zhou et al., 2026a), POMO (Kwon et al., 2020), ELG (Gao et al., 2024), MatNet (Kwon et al., 2021), and Heter-AM (Li et al., 2021); (ii) *generalist neural solvers:* GOAL-MTL (Drakulic et al., 2025), MTPOMO (Liu et al., 2024)¹, MVMoE/4E (abbreviated as MVMoE) (Zhou et al., 2024), ReLD-MTL (Huang et al., 2025), RouteFinder (abbreviated as RF) (Berto et al., 2025), and CaDA (Li et al., 2025a)².

Metrics and Inference We report the optimality gap (Gap) and total inference time (Time) for each method. The gap quantifies the discrepancy between the corresponding solvers and the oracle solver. For neural solvers, we execute the source code provided by the authors using default settings. We compare our URS with relevant baselines for each problem, as well as its single-task version (i.e., URS-STL). The URS-STL variant uses exactly the same architecture and hyperparameters (see Appendix H.2 for more details). For URS, we report the best result with $\times 8$ (Kwon et al., 2020) and $\times 128$ (Kwon et al., 2021) instance augmentation

¹For MTPOMO, we use the implementation in Zhou et al. (2024), ensuring problem consistency across all neural solvers.

²Given the disparities in problem and training settings, detailed comparisons against RF and CaDA are deferred to Appendix J.2.

URS: A Unified Neural Routing Solver for Cross-Problem Zero-Shot Generalization

Table 4. Zero-shot generalization on 1K test instances of 32 unseen VRPs ($N = 100$). Due to page limit, complete experimental results (including all 99 unseen variants) are deferred to Appendix I.

Method	CVRPBP		CVRPBL		MDOCVRPBP		MDOCVRPBTW		MDOCVRPBPL		MDOCVRPBPLTW		MDCVRP		MDOCVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
MVMoE	13.95%	14s	13.38%	14s	56.60%	33s	63.77%	38s	56.26%	34s	62.47%	40s	33.06%	31s	29.08%	31s
MTPOMO	13.71%	7s	13.44%	8s	42.94%	24s	47.09%	29s	41.98%	27s	46.26%	32s	23.38%	23s	27.46%	23s
ReLD-MTL	13.57%	9s	12.96%	10s	37.74%	27s	45.16%	34s	37.45%	29s	44.74%	36s	18.39%	25s	22.32%	25s
URS	12.95%	7s	12.65%	8s	24.44%	22s	26.31%	26s	24.22%	24s	26.20%	28s	15.05%	23s	22.01%	23s

Method	MDCVRPL		MDOCVRPTW		MDOCVRPB		MDOCVRPBL		MDOCVRPLTW		SPCTSP		PDCVRP		OPDCVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	1.6h	0.00%	1.6h	0.00%	1.6h
MVMoE	33.42%	34s	51.57%	40s	28.21%	26s	28.70%	28s	51.09%	43s	—	—	—	—	—	—
MTPOMO	24.32%	25s	35.71%	29s	25.06%	20s	25.01%	21s	35.39%	31s	—	—	—	—	—	—
ReLD-MTL	18.48%	28s	33.55%	33s	18.50%	21s	18.50%	22s	33.26%	35s	—	—	—	—	—	—
URS	15.32%	26s	26.05%	25s	15.17%	18s	14.90%	20s	25.59%	27s	-2.37%	5s	-1.47%	5s	4.93%	5s

Method	ACVRPL		ACVRPBL		ACVRPB		ACVRPBTW		APDTSP		AMDCVRPL		AMDCVRP		APDCVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	-2.24%	1.9m	6.18%	2.4m	7.10%	2.1m	9.95%	2.2m	6.21%	1m	7.15%	6.1m	8.11%	5.3m	7.03%	1.2m

Method	ACVRPLTW		ACVRPBLTW		ACVRPBPLTW		AOCVRPBLTW		ACVRPTW		ACVRPBPTW		AOPDCVRP		AOP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	12.22%	2.7m	10.18%	2.4m	13.82%	2.8m	14.32%	2.5m	11.28%	2.5m	14.52%	2.6m	11.13%	1.2m	-5.47%	1.2m

Table 5. Generalization on different large-scale VRP variants. All models are trained on instances of size $N = 100$ and employ instance augmentation $\times 8$. PyVRP and OR-Tools can process all instances of a given size concurrently to ensure a fair comparison.

Problem	Method	N = 1000		N = 2000		N = 3000		N = 4000		N = 5000	
		Obj.(Gap)	Time	Obj.(Gap)	Time	Obj.(Gap)	Time	Obj.(Gap)	Time	Obj.(Gap)	Time
CVRPTW	PyVRP	159.35(0.00%)	20m	337.15(0.00%)	40m	516.97(0.00%)	1h	618.58(0.00%)	2.7h	787.25(0.00%)	3.3h
	OR-Tools	178.13(11.79%)	20m	357.18(5.94%)	40m	533.97(3.29%)	1h	647.92(4.74%)	2.7h	807.128(2.52%)	3.3h
	MTPOMO	219.22(37.57%)	1.7m	488.40(44.86%)	14.6m	773.00(49.53%)	49.5m	975.61(57.72%)	1.9h	OOM	
	MVMoE	233.53(46.55%)	1.7m	624.73(85.30%)	14.6m	1094.25(111.67%)	51.9m	1486.34(140.28%)	2h	OOM	
	ReLD-MTL	183.31(15.04%)	1.6m	396.28(17.54%)	13.4m	613.58(18.69%)	45.7m	748.65(21.03%)	1.8h	OOM	
	URS	172.58(8.30%)	1.3m	365.27(8.34%)	11.1m	559.03(8.14%)	30.1m	673.63(8.90%)	1.2h	853.79(8.45%)	2.3h
CVRPB	PyVRP	36.52(0.00%)	20m	51.80(0.00%)	40m	75.12(0.00%)	1h	93.52(0.00%)	2.7h	112.21(0.00%)	3.3h
	OR-Tools	44.23(21.11%)	20m	63.33(22.26%)	40m	88.6(17.94%)	1h	108.82(16.36%)	2.7h	128.11(14.17%)	3.3h
	MTPOMO	45.92(25.77%)	1.1m	83.50(61.20%)	8.8m	136.52(81.73%)	31.4m	197.99(111.70%)	1.2h	OOM	
	MVMoE	98.60(170.03%)	1m	250.47(383.57%)	8.1m	332.93(343.19%)	28.9m	423.19(352.49%)	1.1h	OOM	
	ReLD-MTL	38.52(5.48%)	1m	59.21(4.30%)	8m	88.36(17.62%)	28.3m	114.88(22.83%)	1.1h	OOM	
	URS	34.86(-4.54%)	40s	50.39(-2.72%)	5.1m	72.01(-4.15%)	17.3m	91.74(-1.91%)	41.2m	111.93(-0.24%)	1.3h
OCVRPTW	PyVRP	90.91(0.00%)	20m	164.00(0.00%)	40m	224.16(0.00%)	1h	299.45(0.00%)	2.7h	367.86(0.00%)	3.3h
	OR-Tools	105.98(16.58%)	20m	207.041(26.24%)	40m	280.88(25.30%)	1h	370.85(23.84%)	2.7h	447.02(21.52%)	3.3h
	MTPOMO	147.14(61.85%)	1.6m	297.53(81.43%)	12.8m	437.73(95.28%)	44.3m	601.63(100.91%)	1.7h	OOM	
	MVMoE	147.35(62.09%)	1.6m	389.60(137.57%)	12.9m	634.11(182.88%)	44.7m	899.18(200.28%)	1.7h	OOM	
	ReLD-MTL	110.00(21.01%)	1.5m	208.60(27.20%)	12.3m	292.70(30.57%)	42m	394.42(31.72%)	1.6h	OOM	
	URS	98.76(8.63%)	1m	178.07(8.58%)	7.6m	243.88(8.80%)	25.4m	325.64(8.74%)	1h	398.72(8.39%)	1.9h

for symmetric and asymmetric instances, respectively. The mark (—) indicates that the method does not support the corresponding VRP variant.

4.2. Performance Evaluation

Performance on Seen VRPs As presented in Table 2 and Table 3, among comparable neural solvers, URS-STL achieves the lowest gap on 8 of the 11 VRPs while offering fast inference, highlighting the strength of our architecture when fully specialized. Although URS-MTL exhibits a performance degradation relative to STL variants, it remains competitive across all 11 problems. Using ATSP as an exam-

ple, we attribute this drop to geometric structural inequality, which likely biases the model’s learning capacity toward symmetric VRPs. While the lack of explicit constraint states yields weaker performance compared to ReLD-MTL and MVMoE, URS shows clear advantages when addressing unseen VRPs (Table 4) and large-scale instances (Table 5).

Generalization to Unseen VRPs We evaluate our URS on zero-shot generalization across 99 unseen VRP variants. As presented in Table 4 and Appendix I, on 44 widely-studied VRPs with diverse constraint combinations (i.e., C, O, L, TW, MD, B, BP), URS continues to deliver high-

Table 6. Comparison on Set-AGS with $N \in [3000, 7000]$. All models are trained on instances of size $N = 100$.

Type	Method	Leuven1 (N=3000)	Leuven2 (N=4000)	Antwerp1 (N=6000)	Antwerp2 (N=7000)	Avg.gap
Specialist	LEHD greedy	16.60%	34.86%	14.66%	22.77%	22.22%
	BQ greedy	18.53%	30.70%	16.48%	27.67%	23.34%
	POMO aug $\times 8$	460.32%	202.17%	OOM	OOM	—
	ELG aug $\times 8$	12.12%	21.52%	OOM	OOM	—
Generalist	MTPOMO aug $\times 8$	67.72%	87.31%	OOM	OOM	—
	MVMoE/4E aug $\times 8$	299.10%	170.10%	OOM	OOM	—
	MVMoE/4E-L aug $\times 8$	182.28%	127.07%	OOM	OOM	—
	RF-MVMoE aug $\times 8$	57.30%	160.27%	OOM	OOM	—
	RF-TE aug $\times 8$	26.90%	45.56%	OOM	OOM	—
	CaDA aug $\times 8$	1035.51%	OOM	OOM	OOM	—
	ReLD-MTL aug $\times 8$	17.00%	30.59%	OOM	OOM	—
	GOAL greedy	OOM	OOM	OOM	OOM	—
	URS aug $\times 8$	11.50%	17.61%	9.18%	14.86%	13.29%

quality solutions. Notably, URS merely selects the next node from a given set of feasible candidates without any additional domain knowledge, making its performance especially meaningful. Benefiting from our UDR, URS can directly address 55 less-explored VRP variants that compound asymmetry and PD relations. These results highlight URS’s strong capacity for cross-problem zero-shot generalization without any fine-tuning.

Scalability to Larger-scale VRP Variants We conduct experiments on large-scale instances to validate URS’s cross-problem scalability. Specifically, we generate five large-scale (1,000 – 5,000 nodes) synthetic datasets for each variant (CVRPTW, CVRPB, and OCVRPTW), and each dataset comprises 16 instances. We adhere to the capacity constraints specified in TAM (Hou et al., 2022). As shown in Table 5, URS consistently achieves the best solution quality, complemented by remarkably fast inference speeds, across various problem instances among all comparable multi-task neural solvers. Despite a significant increase in the problem scale (up to $50\times$ the training size, $N = 100$), the performance of URS remains stable, with no severe degradation. Remarkably, URS already outperforms the strong PyVRP with a carefully manual design on large-scale CVRPB instances. These extensive results show URS’s strong scalability to large-scale complex VRP variants.

Results on Benchmark Dataset We further evaluate URS on benchmark instances from CVRPLib Set-X (Uchoa et al., 2017) and CVRPLib Set-AGS (also known as Set-XXL) (Arnold et al., 2019). We compare URS against representative specialist and generalist solvers. As presented in Table 6, URS can significantly outperform existing generalist neural solvers in large-scale benchmark instances. Notably, as a generalist solver, URS also greatly surpasses many representative specialist models (e.g., LEHD, BQ, and ELG). In addition, the detailed results on the Set-X dataset (Appendix J.3) show that URS maintains its position as the best-performing overall. These results further underscore the applicability of URS in real-world large-scale scenarios.

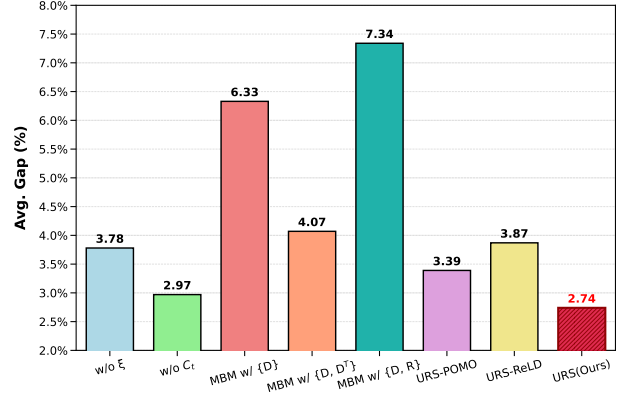


Figure 4. Effects of different components of URS.

Table 7. Results comparing performance with and without MBM, WEIGHT(λ), and BIAS(λ) on seen and unseen variants.

MBM	WEIGHT(λ)	BIAS(λ)	Avg.gap		Best Solution
			Seen (11)	Unseen (11)	
$\times$	$\checkmark$	$\checkmark$	6.33%	10.41%	3/22
$\checkmark$	$\times$	$\times$	4.20%	8.12%	0/22
$\checkmark$	$\times$	$\checkmark$	3.84%	8.02%	1/22
$\checkmark$	$\checkmark$	$\times$	3.63%	7.81%	1/22
$\checkmark$	$\checkmark$	$\checkmark$	2.74%	7.05%	17/22

5. Ablation Study

In this section, we conduct a comprehensive ablation study and analysis to demonstrate the effectiveness and robustness of URS. The detailed results and discussion are presented below. Note that, unless otherwise stated, the instance augmentation technique is used for the ablation studies.

Effects of Key Components We conduct an ablation study across 11 seen VRP variants to validate the effect of key components of URS, mainly including: (1) without node-type indicator ξ of UDR; (2) without problem state C_t ; (3) effects of different priors in MBM; (4) effects of alternative decoder architectures. As detailed in Figure 4, URS maintains its competitive performance without C_t ; its effectiveness is further improved by the inclusion of a node-type indicator ξ . For MBM, the results reveal that three priors $\{D, D^T, R\}$ are complementary, and their combined use allows MBM to effectively learn the geometric and relational biases inherent in various problems, resulting in superior overall performance. Furthermore, replacing the decoder with POMO or ReLD confirms the strong cross-problem robustness of URS, while integrating the decoder introduced in Zhou et al. (2026a) further improves cross-problem performance. More detailed results and analyses can be found in Appendix K.

MBM & Parameter Generator To further evaluate the effects of MBM, WEIGHT(λ), and BIAS(λ) for cross-

Table 8. Performance comparison of ablating each UDR component (i.e., ρ , ω , and ξ) on seen and unseen variants. Since removing positional identifiers ρ prevents the use of instance augmentation, this augmentation technique is disabled across all results to maintain a consistent inference strategy.

ρ	ω	ξ	Avg. gap		Best Solution
			Seen (11)	Unseen (11)	
×	✓	✓	4.65%	18.12%	3/22
✓	×	✓	9.81%	18.60%	0/22
✓	✓	×	5.43%	18.60%	4/22
✓	✓	✓	4.17%	8.51%	15/22

problem zero-shot generalization, we conduct tests on 11 seen problems and 11 unseen CVRP variants widely studied in (Liu et al., 2024). As shown in Table 7, the full URS model achieves the lowest average gap on both seen and unseen problems, and provides the best solution on the majority of tasks (17/22). This shows that their synergy is essential for robust cross-problem zero-shot generalization. We provide detailed results in Appendix K.5.

Effects of Each UDR Component To clarify the independent contribution of each UDR component, we conduct ablations by removing each component in turn. As shown in Table 8, the absence of any single component leads to notable performance degradation. A complete UDR achieves the lowest average gap, affirming the necessity of our UDR design for robust zero-shot generalization. The detailed results for each VRP variant can be found in Appendix K.6.

Hypernetwork vs. Adapter We train URS-Adapter, a URS variant that adopts a distinct decoder for each of the 11 training problems, and compare it with the original URS with a hypernetwork on the same 11 seen problems and large-scale CVRP variants (see Appendix K.7 for detailed results). URS consistently outperforms URS-Adapter across the 11 problems seen, achieving higher parameter efficiency. This performance advantage holds consistently across large-scale CVRP variants, further solidifying our conclusion.

6. Conclusion, Limitation, and Future Work

In this work, we propose a novel RL-based URS for cross-problem zero-shot generalization. The core of URS is the proposed UDR, which broadens problem coverage by substituting problem enumeration with data unification. Its generalization capability is further enhanced by an MBM and a problem-conditioned parameter generator. Extensive experiments show that URS can produce high-quality solutions for 110 VRP variants without any fine-tuning, including 99 unseen variants, significantly broadening the problem coverage and reducing reliance on domain expertise.

Limitation and Future Work While URS achieves impressive results across a wide range of VRP variants, its current training scheme does not fully account for the inherent disparities in problem complexity. In the future, we aim to explore more effective strategies, such as hierarchical training and difficulty-adaptive sampling, to improve training efficiency. Additionally, extending our method to accommodate more complex variants (e.g., dynamic VRP) is a promising direction for future research.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Related Work

A.1. Neural Routing Solvers with Single-Task Learning

The NCO methods have emerged as a promising paradigm for tackling various routing problems, achieving impressive empirical performance. Early seminal work based on Pointer Networks ignited interest in learning end-to-end construction policies for TSP and CVRP (Vinyals et al., 2015; Bello et al., 2016; Nazari et al., 2018). Subsequently, the Transformer-style heavy-encoder and light-decoder auto-regressive architecture has become the dominant paradigm (Kool et al., 2019), while POMO (Kwon et al., 2020) exploits permutation symmetry to enhance exploration diversity. Many advancements within this paradigm have pushed promising results on small instances (e.g., 100-node TSPs) (Xin et al., 2020; Kim et al., 2022; Xiao et al., 2024a;b; Sun et al., 2024). However, models trained on a fixed instance scale often degrade sharply when evaluated out of scale. To mitigate the poor generalization performance, recent studies introduce auxiliary local policies (Gao et al., 2024), adaptive perception across instance scales (Zhou et al., 2026a), or different search space reduction methods (Fang et al., 2024; Zhou et al., 2026b; Chen et al., 2025). In parallel, "heavy decoder" designs that dynamically recompute embeddings at each construction step (Drakulic et al., 2023; Luo et al., 2023) demonstrate strong generalization on large-scale instances. To improve the performance on large-scale routing instances, complementary directions are reducing solving difficulty via problem decomposition (Zheng et al., 2024b; Ye et al., 2024; Li et al., 2025b) or adopting non-autoregressive generation augmented with additional searches (e.g., Monte Carlo tree search (MCTS) and 2-opt) (Fu et al., 2021; Sun & Yang, 2023; Qiu et al., 2022; Li et al., 2023; Ye et al., 2023). Moreover, heterogeneous vehicle routing (Li et al., 2022), asymmetric distance settings (Kwon et al., 2021), multiple VRPs (Zheng et al., 2024a), and pickup-delivery problems (Li et al., 2021; Kong et al., 2024) have each motivated tailored neural designs, which have yielded competitive results. Despite these advancements, they still train a specialist model per problem type, limiting cross-problem transfer. We instead focus on a unified neural routing solver that aims to generalize across multiple routing problems without retraining from scratch for each specification.

A.2. Neural Routing Solvers with Multi-Task Learning

To address the challenge of cross-problem generalization, growing attention has shifted to multi-task learning capable of adapting to diverse routing problems. Existing efforts can broadly fall into two categories: (1) constraint combination-based multi-task learning and (2) adapter-based fine-tuning frameworks. The former line (e.g., MTPOMO (Liu et al., 2024)) treats VRP variants as combinations of a predefined set of attributes and trains a shared backbone, achieving promising results on up to 16 VRP variants. Follow-up work extends this paradigm by introducing mixture-of-experts (MoE) modules (Zhou et al., 2024), heavy decoder designs plus knowledge distillation (Zheng et al., 2025), and constraint-aware dual-attention (Li et al., 2025a) or updated Transformer structure (Berto et al., 2025) to improve model performance. While this enables knowledge sharing across problems, its coverage is inherently bounded by the hand-specified attribute set. The second family trains a shared backbone and performs adaptation to each variant via lightweight adapters. For instance, Lin et al. (2024) pretrain a TSP backbone and adapt to new variants through adapter-based fine-tuning. Drakulic et al. (2025) integrate edge features and adopt supervised multi-task pretraining for obtaining a powerful backbone before fine-tuning on novel tasks. Wang et al. (2025a) propose a multi-armed bandit strategy to realize more efficient multi-task training. Although these reduce retraining cost, they fail to achieve zero-shot generalization. In contrast, we focus on a constructive neural routing solver supported with zero-shot capability: once trained, it can directly generate high-quality solutions for previously unseen problems without any fine-tuning.

B. Setups of VRP Variants

In this paper, we evaluate URS across 110 VRP variants spanning 10 distinct tags, in which each variant may involve either a single tag or a combination of the following tags: (1) Capacity (C); (2) Open Route (O); (3) Backhaul (B); (4) Backhaul and Priority (BP); (5) Duration Limit (L); (6) Time Windows (TW); (7) Multi-Depot (MD); (8) Prize Collecting (PC); (9) Asymmetric (A); (10) Pickup and Delivery (PD). The node coordinates are randomly sampled from the unit square when available (in symmetric geometric cases, i.e., $d_{ij} = d_{ji}$), following AM (Kool et al., 2019). Below, we provide a comprehensive description of the data generation process for each tag.

Capacity (C) The selection of the vehicle capacity C is critical, as prior work (Luo et al., 2025c) demonstrates its significant impact on the solution structure. For PDVRP variants (e.g., PDCVRP, OPDCVRP, APDCVRP, and AOPDCVRP), an excessively large C causes the solution to approximate that of a PDTSP instance. Conversely, an overly restrictive C promotes a simplistic, repetitive Pickup-Delivery sequence. To ensure structurally sound solutions, we set $C=20$ for PDCVRP variants. For all other VRP variants, we adopt the convention from AM (Kool et al., 2019), setting $C=50$.

Open Route (O) The open route setting implies that vehicles are not required to return to the depot after completing their service tours. This does not alter the data generation, and it specifically affects the final cost evaluation by excluding the travel distance from the last customer back to the depot.

Backhaul (B) The standard CVRP involves vehicles delivering goods from a depot to a set of customers with positive demands. The introduction of backhauls extends this formulation to the CVRP with Backhauls (CVRPB), where vehicles must also load goods from backhaul customers and return them to the depot, in addition to serving the standard linehaul (delivery) customers. Following the experimental setup of MTPOMO (Liu et al., 2024), we generate customer demands by sampling from a discrete uniform distribution over the integers $\{1, \dots, 9\}$. Subsequently, 20% of these nodes are randomly designated as backhaul customers, and their corresponding demand values are negative, while the remainder are classified as linehaul customers. Notably, no precedence constraints are imposed between the servicing of linehaul and backhaul customers. To ensure the feasibility of the solutions constructed by our neural solver, we adhere to the MVMoE (Zhou et al., 2024) framework’s policy that the first customer visited must be a linehaul node.

Backhaul and Priority (BP) This constraint imposes a strict precedence requirement on the standard CVRPB, dictating that all linehaul customers must be serviced prior to any backhaul customers. Consequently, once a vehicle serves a backhaul customer, it is prohibited from visiting any subsequent linehaul nodes within the same route. If the first customer served is a backhaul node, the vehicle’s load is initialized to zero. This ensures the neural solver can construct a feasible solution.

Duration Limit (L) Following MTPOMO (Liu et al., 2024), we impose a duration limit (i.e., the maximum length of each vehicle route) of 3 in symmetric cases. Since the node coordinates are randomly sampled from the unit square, the maximum possible distance between any two nodes is $\sqrt{2}$. The setting is sufficient to ensure a vehicle can deliver goods to at least one customer and return to the depot, thus guaranteeing the feasibility of the solution. For the setting of asymmetric cases, we provide details in the description of the following asymmetric instances.

Time Windows (TW) Consistent with the MVMoE (Zhou et al., 2024), we set time windows of the depot to $[e_0, l_0]=[0,3]$ and its service time to 0. The service time s_i for each customer node is set to 0.2. For the time windows for each customer node, we generate them following the methodology outlined in MVMoE (Zhou et al., 2024).

Multi-Depot (MD) The generation of coordinates for the multiple depots aligns with that of other nodes, with the number of depots set to 3 (Berto et al., 2025). A vehicle is required to return to the same depot it departed from. To guarantee that each starting node is considered in this multi-depot variant, we set the number of trajectories for an instance to be the product of the number of depots and the number of customer nodes, maximizing the exploration of high-quality solutions.

Prize Collecting (PC) Under the Prize Collecting condition, each node (excluding the depot) is assigned both a prize and a penalty. A prize is collected for each visited node, while a penalty is incurred for each unvisited node. The objective is to minimize the total travel distance plus the sum of penalties for all unvisited nodes. A key constraint is that a minimum amount of total prize must be collected before the vehicle is permitted to return to the depot and terminate its route. In this context, the minimum prize requirement for the four examined problems (PCTSP, SPCTSP, APCTSP, and ASPCTSP) is

standardized to 1.0. Consistent with the AM (Kool et al., 2019) framework, the prizes and penalties are generated as follows: $prize \sim \text{Uniform}(0, \frac{4}{n})$, and $penalty \sim \text{Uniform}(0, 3 \cdot \frac{k_n}{n})$, where $n = 100$ and $k_n = 4$ in our experiments. For the OP, the objective is to maximize collected prizes without exceeding a predefined distance limit (set to 4.0 for OP and 1.0 for AOP), where unvisited nodes carry no penalties. The prize for each node is assigned proportional to its Euclidean distance from the depot. By normalizing distances to the instance’s maximum distance and discretizing them into $[0.01, 1.00]$, nodes further from the depot systematically yield higher rewards, with the depot’s prize set to zero.

Asymmetric (A) We adopt the data generation method from MatNet (Kwon et al., 2021) to create an asymmetric distance matrix. Due to the small pairwise distances in the generated distance matrix, the parameters for duration limit and time windows set for symmetric cases are not applicable to the asymmetric problems. To ensure the constraints remain valid, we adjust the duration limit and depot end time to 0.6 and 1.0, respectively.

Pickup and Delivery (PD) For PDP variants, we adhere to the methodology in Li et al. (2021). For a problem of size N , node 0 is set to the depot, nodes 1 to $N/2$ are designated as pickup nodes, while nodes from $N/2 + 1$ to N are designated as delivery nodes. For each pickup node v_i , its corresponding delivery node is $v_{i+N/2}$. A strict precedence constraint is imposed in PDP cases, requiring each vehicle to visit a pickup node before its corresponding delivery node. For PDCVRP variants, the demand for each delivery node is randomly sampled from a discrete uniform distribution over $\{1, 2, \dots, 9\}$. The demand for each pickup node is then set to the negative of its corresponding delivery node’s demand. To ensure the feasibility of the constraints, the vehicle capacity is set to 20.

C. Unified Data Representation

In this section, we provide a detailed description of our UDR $U = \{\mathbf{u}_i\}_{i=0}^n$ (see Table 9). Each node $\mathbf{u}_i$ consists of three components, i.e., $\mathbf{u}_i = \{\rho_i, \omega_i, \xi_i\}$, where ρ_i , ω_i , ξ_i are a positional identifier, unified attribute set, and node-type indicator, respectively. To distinguish linehaul/delivery from backhaul/pickup nodes, we assign negative demand values to backhaul/pickup nodes (positive for linehaul/delivery), making demand the only signed component, following (Liu et al., 2024; Zhou et al., 2024).

Table 9. Detailed description of each component in UDR.

Attribute Name	Symbol	Range	Description	Examples
Random Identifier	η	$[0, 1]$	Random scalar for asymmetric problems.	ATSP
Node Coordinates	$\{x, y\}$	$[0, 1]^2$	2D coordinates of the node.	TSP, CVRP
Demand	δ	$[-1, 1]$	Node demand (+ for delivery, – for pickup).	CVRP, PDCVRP
Prize	ϵ	$[0, 1]$	Prize collected for each node.	OP, PCTSP
Penalty	μ	$[0, 1]$	Penalty incurred for each node.	PCTSP
Earliest Arrival Time	e	$[0, 1]$	Earliest permissible arrival time.	CVRPTW
Latest Arrival Time	l	$[0, 1]$	Latest permissible arrival time (deadline).	CVRPTW
Service Time	s	$[0, 1]$	Time required for service at the node.	CVRPTW
Depot	–	$\{0, 1\}$	Indicates if the node is a depot.	CVRP
Pickup Node	–	$\{0, 1\}$	Indicates if the node is a pickup location.	PDTSP
Delivery Node	–	$\{0, 1\}$	Indicates if the node is a delivery location.	CVRP, PDTSP
Node in Sub-routes	–	$\{0, 1\}$	The solution π can have sub-routes.	CVRP
Node in Open Route	–	$\{0, 1\}$	Vehicles need not return to the depot.	OCVRP

D. Multi-hot Problem Representation

Leveraging the UDR, each problem instance activates only a subset of the unified feature space. Thus, for any instance $\mathcal{G}_i$ from different problems, we simply mark active (non-zero) feature slots with 1 to obtain a corresponding multi-hot problem representation λ_i . Different from raw feature values, it encodes presence only and provides conditional signals to the following position bias weight (see Equation (6)) and adaptive decoder (see Equation (9) and Equation (10)), helping the model distinguish variants and refine node selection. Here, we provide detailed representations for each problem in our experiments, and each problem is a 13-dimensional multi-hot vector.

As shown in Table 10, we train URS on eleven classic and varied routing problems to ensure that each feature in the UDR has been seen at least once, i.e., $\mathcal{P} = \{P_i\}_{i=1}^{11}$. We adhere to the principle of minimal redundancy and strictly align with existing work on CVRP variant selection (e.g., MVMoE (Zhou et al., 2024) and ReLD (Huang et al., 2025)): most features in UDR appear in at least two training problems, helping prevent URS from overfitting to a single problem. The only exception is the Penalty attribute, as there is structural similarity between the seen PCTSP and unseen SPCTSP. For more detailed analysis and experimental results, please see Appendix J.1.

Then, we evaluate generalization on unseen problems on different combinations of ten constraint attributes (see Appendix B), and the detailed representations of each unseen problem can be found in Table 11 (unseen well-studied VRP variants) and Table 12 (unseen less-explored VRP variants).

Table 10. The representations for seen routing problems in our experiments. The total number of problems is 11. In training, each feature in the UDR has been seen at least once. Here, RI represents the random identifier, and EAT, LAT, and ST indicate the earliest arrival time, the latest arrival time, and the service time, respectively.

Problem	Oracle	RI	Coord.	Demand	Prize	Penalty	EAT	LAT	ST	Depot	Pickup	Delivery	Sub-routes	Open Route
ATSP	LKH-3	✓												
TSP	LKH-3		✓											
OP	Compass		✓		✓					✓				
PCTSP	ILS		✓		✓	✓				✓				
PDTSP	LKH-3		✓							✓	✓	✓		
ACVRP	PyVRP	✓		✓						✓		✓	✓	
CVRP	PyVRP		✓	✓						✓		✓	✓	
CVRPTW	PyVRP		✓	✓			✓	✓	✓	✓		✓	✓	
CVRPB	OR-Tools		✓	✓						✓	✓	✓	✓	
OCVRP	LKH-3		✓	✓						✓		✓	✓	✓
OCVRPTW	OR-Tools		✓	✓			✓	✓	✓	✓		✓	✓	✓
Frequency		2/11	9/11	6/11	2/11	1/11	2/11	2/11	2/11	9/11	2/11	7/11	6/11	2/11

Table 11. The problem representations of unseen 44 well-studied VRP variants. Here, RI represents the random identifier, and EAT, LAT, and ST indicate the earliest arrival time, the latest arrival time, and the service time, respectively.

Problem	Oracle	RI	Coord.	Demand	Prize	Penalty	EAT	LAT	ST	Depot	Pickup	Delivery	Sub-routes	Open Route
CVRPL	LKH-3		✓	✓						✓		✓	✓	
OCVRPB	OR-Tools		✓	✓						✓	✓	✓	✓	✓
CVRPBL	OR-Tools		✓	✓						✓	✓	✓	✓	
CVRPLTW	OR-Tools		✓	✓			✓	✓	✓	✓		✓	✓	
OCVRPBTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
CVRPBLTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	
OCVRPL	OR-Tools		✓	✓						✓		✓	✓	✓
CVRPBTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	
OCVRPBL	OR-Tools		✓	✓						✓	✓	✓	✓	✓
OCVRPLTW	OR-Tools		✓	✓			✓	✓	✓	✓		✓	✓	✓
OCVRPBLTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
CVRPBP	OR-Tools		✓	✓						✓	✓	✓	✓	
OCVRPBP	OR-Tools		✓	✓						✓	✓	✓	✓	✓
CVRPBPL	OR-Tools		✓	✓						✓	✓	✓	✓	
OCVRPBPTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
CVRPBPLTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	
CVRPBPTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	
OCVRPBPL	OR-Tools		✓	✓						✓	✓	✓	✓	✓
OCVRPBPLTW	OR-Tools		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
MDCVRP	PyVRP		✓	✓						✓		✓	✓	
MDCVRPTW	PyVRP		✓	✓			✓	✓	✓	✓		✓	✓	
MDOCVRP	PyVRP		✓	✓						✓		✓	✓	✓
MDCVRPL	PyVRP		✓	✓						✓		✓	✓	
MDCVRPB	PyVRP		✓	✓						✓	✓	✓	✓	
MDOCVRPTW	PyVRP		✓	✓			✓	✓	✓	✓		✓	✓	✓
MDOCVRPB	PyVRP		✓	✓						✓	✓	✓	✓	✓
MDCVRPBL	PyVRP		✓	✓						✓	✓	✓	✓	
MDCVRPLTW	PyVRP		✓	✓			✓	✓	✓	✓		✓	✓	
MDOCVRPBTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
MDCVRPBLTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	
MDOCVRPL	PyVRP		✓	✓						✓		✓	✓	✓
MDCVRPBTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	
MDOCVRPBL	PyVRP		✓	✓						✓	✓	✓	✓	✓
MDOCVRPLTW	PyVRP		✓	✓			✓	✓	✓	✓		✓	✓	✓
MDOCVRPBLTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
MDCVRPBP	PyVRP		✓	✓						✓	✓	✓	✓	
MDOCVRPBP	PyVRP		✓	✓						✓	✓	✓	✓	✓
MDCVRPBPL	PyVRP		✓	✓						✓	✓	✓	✓	
MDOCVRPBPTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
MDCVRPBPLTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	
MDCVRPBPTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	
MDOCVRPBPL	PyVRP		✓	✓						✓	✓	✓	✓	✓
MDOCVRPBPLTW	PyVRP		✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
SPCTSP	ILS		✓		✓	✓				✓				

Table 12. The problem representations of unseen 55 less-explored VRP variants. Here, RI represents the random identifier, and EAT, LAT, and ST indicate the earliest arrival time, the latest arrival time, and the service time, respectively.

Problem	Oracle	RI	Coord.	Demand	Prize	Penalty	EAT	LAT	ST	Depot	Pickup	Delivery	Sub-routes	Open Route
ACVRPTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	
AOCVRP	OR-Tools	✓		✓						✓		✓	✓	✓
ACVRPL	OR-Tools	✓		✓						✓		✓	✓	
ACVRPB	OR-Tools	✓		✓						✓	✓	✓	✓	
AOCVRPTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	✓
AOCVRPB	OR-Tools	✓		✓						✓	✓	✓	✓	✓
ACVRPBL	OR-Tools	✓		✓						✓	✓	✓	✓	
ACVRPLTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	
AOCVRPBTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	✓
ACVRPBLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
AOCVRPL	OR-Tools	✓		✓						✓		✓	✓	✓
ACVRPBTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
AOCVRPBL	OR-Tools	✓		✓						✓	✓	✓	✓	✓
AOCVRPLTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	✓
AOCVRPBLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓
ACVRPBP	OR-Tools	✓		✓						✓	✓	✓	✓	
AOCVRPBP	OR-Tools	✓		✓						✓	✓	✓	✓	✓
ACVRPBPL	OR-Tools	✓		✓						✓	✓	✓	✓	
AOCVRPBPTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓
ACVRPBPLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
ACVRPBPTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
AOCVRPBPL	OR-Tools	✓		✓						✓	✓	✓	✓	✓
AOCVRPBPLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓
AMDCVRP	OR-Tools	✓		✓						✓		✓	✓	
AMDCVRPTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	
AMDOCVRP	OR-Tools	✓		✓						✓		✓	✓	✓
AMDCVRPL	OR-Tools	✓		✓						✓		✓	✓	
AMDCVRPB	OR-Tools	✓		✓						✓	✓	✓	✓	
AMDOCVRPTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	✓
AMDOCVRPB	OR-Tools	✓		✓						✓	✓	✓	✓	✓
AMDCVRPBL	OR-Tools	✓		✓						✓	✓	✓	✓	
AMDCVRPLTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	
AMDOCVRPBTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓
AMDCVRPBLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
AMDOCVRPL	OR-Tools	✓		✓						✓		✓	✓	✓
AMDCVRPBTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
AMDOCVRPBL	OR-Tools	✓		✓						✓	✓	✓	✓	✓
AMDOCVRPLTW	OR-Tools	✓		✓			✓	✓	✓	✓		✓	✓	✓
AMDOCVRPBLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓
AMDCVRPBP	OR-Tools	✓		✓						✓	✓	✓	✓	
AMDOCVRPBP	OR-Tools	✓		✓						✓	✓	✓	✓	✓
AMDCVRPBPL	OR-Tools	✓		✓						✓	✓	✓	✓	
AMDOCVRPBPTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓
AMDCVRPBPLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
AMDCVRPBPTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	
AMDOCVRPBPL	OR-Tools	✓		✓						✓	✓	✓	✓	✓
AMDOCVRPBPLTW	OR-Tools	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓
APDTSP	OR-Tools	✓								✓	✓	✓		
APDCVRP	OR-Tools	✓		✓						✓	✓	✓	✓	
AOPDCVRP	OR-Tools	✓		✓						✓	✓	✓	✓	✓
PDCVRP	OR-Tools		✓	✓						✓	✓	✓	✓	
OPDCVRP	OR-Tools		✓	✓						✓	✓	✓	✓	✓
AOP	OR-Tools	✓			✓					✓				
APCTSP	OR-Tools	✓			✓	✓				✓				
ASPCTSP	N/A	✓			✓	✓				✓				

E. Model Architecture and Training

E.1. Encoder

Similar to previous work (Kool et al., 2019; Kwon et al., 2020), we also use the encoder composed of L stacked attention layers, where each attention layer consists of two sub-layers: an attention sub-layer and a Feed-Forward (FF) sub-layer, both of which use Instance Normalization (Ulyanov et al., 2016) and skip-connection (He et al., 2016).

The encoder θ_{enc} with L stacked attention layers transforms $H^{(0)}$ into advanced node embeddings $H^{(L)} = \{\mathbf{h}_i^{(L)}\}_{i=0}^n$. Let $H^{(\ell-1)} = \{\mathbf{h}_i^{(\ell-1)}\}_{i=0}^n$ denote the input to the ℓ -th attention layer for $\ell = 1, \dots, L$. The outputs for the i -th node are computed as follows:

$$\tilde{\mathbf{h}}_i^{(\ell)} = \text{IN}^{(\ell)} \left(\mathbf{h}_i^{(\ell-1)} + \text{MBM}^{(\ell)} \left(\mathbf{h}_i^{(\ell-1)}, H^{(\ell-1)}, \mathbf{D}, \mathbf{D}^T, \mathbf{R} \right) \right), \quad (11)$$

$$\mathbf{h}_i^{(\ell)} = \text{IN}^{(\ell)} \left(\tilde{\mathbf{h}}_i^{(\ell)} + \text{FF}^{(\ell)} \left(\tilde{\mathbf{h}}_i^{(\ell)} \right) \right), \quad (12)$$

where $\text{IN}(\cdot)$ denotes instance normalization (Ulyanov et al., 2016), MBM represents the adopted mixed bias module (see Equation (5) for details), and $\text{FF}(\cdot)$ in Equation (12) corresponds to a fully connected neural network with ReLU activation. After L attention layers, the final node embeddings $H^{(L)} = \{\mathbf{h}_i^{(L)}\}_{i=0}^n$ encapsulate the advanced feature representations of each node. Notably, to ensure dimensional consistency across bias channels, we normalize $\mathbf{D}$ and $\mathbf{D}^T$ by dividing their maximum value before passing the MBM.

E.2. Decoder

In URS, all parameters $\theta_{dec}(\lambda)$ are adaptively generated conditioned on the multi-hot problem representation λ (see Appendix D for detailed problem representations). $\theta_{dec}(\lambda)$ includes two MLP networks, which are WEIGHT(λ) and BIAS(λ). WEIGHT(λ) is used to generate linear projection matrices $[W_Q(\lambda), W_K(\lambda), W_V(\lambda)]$. BIAS(λ) in Equation (6) is used to generate bias weights for Attention($\cdot$) (see Equation (9)) and compatibility (see Equation (10)).

For WEIGHT(λ), we use a multi-layer MLP hypernetwork conditioned on a 13-dimensional problem representation λ to generate problem-specific adaptive parameters. Note that we factor $W_Q(\lambda) \in \mathbb{R}^{(2d+1) \times d}$ into three vertically concatenated blocks $W_{\text{first}}(\lambda) \in \mathbb{R}^{d \times d}$, $W_{\text{last}}(\lambda) \in \mathbb{R}^{d \times d}$ and $W_C(\lambda) \in \mathbb{R}^{1 \times d}$, applied respectively to the first visited node $\mathbf{h}_{\pi_1}^{(L)}$, the last visited node $\mathbf{h}_{\pi_{t-1}}^{(L)}$, and a scalar state feature $C_t \in \mathbb{R}^1$ (see Appendix E.3 for more details). Thus WEIGHT(λ) outputs the linear projection matrices $\{W_{\text{first}}(\lambda), W_{\text{last}}(\lambda), W_C(\lambda), W_K(\lambda), W_V(\lambda)\}$.

Furthermore, we adopt the parameter compression technique of Ha et al. (2016) to control model size. Specifically, we first generate an advanced embedding matrix $\mathbf{H}_{\text{dec}}(\lambda) = [\mathbf{h}_{\text{dec}}^i(\lambda) | i = 1, 2, \dots, 5] \in \mathbb{R}^{5 \times |\lambda|}$ whose five row embeddings each parameterize one target projection matrix. It can be computed via three consecutive linear projections:

$$\mathbf{H}_{\text{dec}}(\lambda) = ((\lambda W_1 + \mathbf{b}_1) W_2 + \mathbf{b}_2) W_3 + \mathbf{b}_3, \quad (13)$$

where $W_1 \in \mathbb{R}^{|\lambda| \times d_h}$, $W_2 \in \mathbb{R}^{d_h \times d_h}$, $W_3 \in \mathbb{R}^{d_h \times (5 \times |\lambda|)}$, $\mathbf{b}_1 \in \mathbb{R}^{d_h}$, $\mathbf{b}_2 \in \mathbb{R}^{d_h}$, $\mathbf{b}_3 \in \mathbb{R}^{(5 \times |\lambda|)}$ are learnable parameters, d_h is 256 in this work, following Lin et al. (2022). Next, we use five different linear projections to map the new representation $\mathbf{h}_{\text{dec}}^i(\lambda)$ of each row to the corresponding decoder parameters. The computation can be expressed as:

$$W_{\text{first}}(\lambda) = \mathbf{h}_{\text{dec}}^1(\lambda) W_{\text{hyper}}^1, \quad W_{\text{last}}(\lambda) = \mathbf{h}_{\text{dec}}^2(\lambda) W_{\text{hyper}}^2, \quad W_C(\lambda) = \mathbf{h}_{\text{dec}}^3(\lambda) W_{\text{hyper}}^3, \quad (14)$$

$$W_K(\lambda) = \mathbf{h}_{\text{dec}}^4(\lambda) W_{\text{hyper}}^4, \quad W_V(\lambda) = \mathbf{h}_{\text{dec}}^5(\lambda) W_{\text{hyper}}^5, \quad (15)$$

where $W_{\text{hyper}}^1 \in \mathbb{R}^{|\lambda| \times (d \times d)}$, $W_{\text{hyper}}^2 \in \mathbb{R}^{|\lambda| \times (d \times d)}$, $W_{\text{hyper}}^3 \in \mathbb{R}^{|\lambda| \times (1 \times d)}$, $W_{\text{hyper}}^4 \in \mathbb{R}^{|\lambda| \times (d \times d)}$, $W_{\text{hyper}}^5 \in \mathbb{R}^{|\lambda| \times (d \times d)}$ are learnable parameters. In this way, we use a simple MLP hypernetwork WEIGHT(λ) to generate used linear projection matrices $\{W_Q(\lambda), W_K(\lambda), W_V(\lambda)\}$ conditioned on the multi-hot problem representation λ , enabling the adaptive construction of high-quality solutions for different problems.

E.3. Problem State C_t

In this section, we detail the problems that require additional state signals. If a problem is not mentioned in this section, it means we will not apply any extra state to it, i.e., $C_t = 0$.

For all problems with the capacity constraint, we explicitly impose **only remaining load**, while diverse additional constraints (e.g., time windows, duration limit, and backhaul) are enforced implicitly as $\mathcal{M}_t$ masks infeasible candidate nodes to guarantee valid solutions, instead of being fully enumerated in the input of the decoder. For OP and AOP, we keep track of the remaining maximum length at time t , and C_t represents the remaining routing length that can be moved, and we normalize it to $[0, 1]$ by dividing by the maximum length (i.e., 4.0 in OP100 and 1.0 in AOP100). The setting is the same as AM (Kool et al., 2019). For (A)PCTSP and (A)SPCTSP, C_t is the remaining prize to collect, and we do not provide any information about the penalties, as this is irrelevant for the remaining decisions, following Kool et al. (2019). For the problem state, we provide an ablation study in Appendix K.2.

E.4. Training

Following Kwon et al. (2020), we use n trajectories with distinct starting nodes in training. URS is trained by the REINFORCE (Williams, 1992) algorithm with a shared baseline (Kwon et al., 2020):

$$\nabla_{\theta} \mathcal{L}(\theta) = \mathbb{E}_{p(\pi|\mathcal{G}, \theta)} \left[(f(\pi|\mathcal{G}) - \frac{1}{n} \sum_{i=1}^n f(\pi^i|\mathcal{G})) \nabla_{\theta} \log p_{\theta}(\pi|\mathcal{G}) \right], \quad (16)$$

$$p_{\theta}(\pi | \mathcal{G}) = \prod_{t=2}^n p_{\theta}(\pi_t | \mathcal{G}, \pi_{1:t-1}), \quad (17)$$

where the objective function $f(\pi|\mathcal{G})$ represents the total reward (e.g., the negative value of tour length) of instance $\mathcal{G}$ given a specific solution π .

F. Adaptation Attention Free Module

Following Zhou et al. (2026a), we implement Attention($\cdot$) using an adaptation attention free module (AAFM) to enhance geometric pattern recognition for routing problems. Given the input X , the AAFM computation is then expressed as:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V, \quad (18)$$

$$\text{Attention}(Q, K, V, A) = \sigma(Q) \odot \frac{\exp(A)(\exp(K) \odot V)}{\exp(A) \exp(K)}, \quad (19)$$

where W^Q , W^K , and W^V are learnable matrices. σ denotes the sigmoid function, $\odot$ represents the element-wise product, and $A = \{a_{ij}\}$ denotes the pair-wise adaptation bias. For distance matrices, the corresponding adaptation bias $f(\alpha, N, d_{ij}) = -\alpha \cdot \log_2 N \cdot d_{i,j}$. Here, N denotes the total number of nodes (i.e., problem size), and α is generated via a lightweight network BIAS(λ) in our paper (see Equation (6)). Notably, for the relation matrix, we remove the scale N in the calculation of adaptation bias because the relation is scale-independent. Compared to multi-head attention (MHA) (Vaswani et al., 2017), AAFM enables the model to explicitly capture instance-specific knowledge by updating pair-wise adaptation biases while exhibiting lower computational overhead. Further details are provided in the related work mentioned above.

G. Comparison between Different Attention Mechanisms

Table 13. Comparison between different attention mechanisms. Here, N, d, m denote the sequence length, feature dimension, and extra hidden layer dimension of mixed-score attention, respectively. Values in parentheses represent the percentage increase or decrease relative to URS and Avg. Peak VRAM represents the average peak VRAM per instance.

Attention Mechanism	Time Complexity	Space Complexity	#Params	Avg. Peak VRAM	Inference Time
Multi-Head Attention	$\mathcal{O}(N^2 d)$	$\mathcal{O}(N^2 + Nd)$	3.32M($\times 0.7$)	1.87MB($\times 0.99$)	0.92s($\times 0.9$)
Mixed-Score Attention (Parallel)	$\mathcal{O}(N^2(d + m))$	$\mathcal{O}(N^2 m + Nd)$	16.17M($\times 3.3$)	25.64MB($\times 13.6$)	3.64s($\times 3.7$)
Mixed Bias Module (Ours)	$\mathcal{O}(N^2 d)$	$\mathcal{O}(N^2 + Nd)$	4.96M	1.89MB	0.97s

To validate the efficiency of our proposed MBM, we have conducted a new comparative experiment between different attentions and tested them on TSP instances to ensure consistent decoding steps. Specifically, we replace our MBM with both a vanilla MHA (Vaswani et al., 2017) and the parallel mixed-score attention (MSA) used in MatNet (Kwon et al., 2021),

and compare their complexities, memory overheads, and inference time. The results in Table 13 show that the increased VRAM and time overhead of MBM compared to standard MHA are small, and both maintain the same time complexity and space complexity. Compared to MatNet’s parallel MSA, our MBM drastically reduces both VRAM consumption and inference latency. These results show that MBM imposes only modest computational penalties while successfully capturing crucial problem-specific biases, which enhances cross-problem generalization while reducing architectural redundancy.

H. Training and Model Settings

H.1. Multi-Task Version

The detailed information about the hyperparameter settings can be found in Table 14. The URS training is conducted on a single NVIDIA GeForce RTX 4090 GPU (24GB of memory). The training procedure requires only about 95 hours (≈ 4 days) in total (500 epochs at approximately 12 minutes each) to obtain a unified parameter set capable of addressing a broad class of routing problems.

H.2. Single-Task Version

URS-STL is trained on data from only a single problem type (e.g., "URS-STL (CVRP)" is trained only on CVRP). The URS-STL variant uses exactly the same model architecture, hyperparameters (e.g., embedding dimension, number of layers, optimizer), and training setup (e.g., initial learning rate, batch size) as the multi-task URS described in Section 4.1 and Appendix H.1. The only modification we made is to the training duration. We observe that the single-task models converge significantly faster than the multi-task model. Therefore, we train URS-STL models for 300 epochs, with a learning rate decay applied at the 251st epoch. This contrasts with the multi-task URS, which is trained for 500 epochs with decay applied at the 451st epoch. This adjustment is made solely to reflect the faster convergence of the single-task training. All other hyperparameters and model architecture remain identical.

Table 14. The hyperparameter settings of URS.

Hyperparameter	Value
Optimizer	AdamW
Clipping parameter ζ	50
Initial learning rate	10^{-4}
Epochs of learning rate decay	[451]
Factor of learning rate decay	0.1
Weight decay	10^{-6}
The number of encoder layers L	12
Embedding dimension d	128
Feed forward dimension	512
Dimension d_h in Equation (13)	256
Training scale	100
Batches of each epoch	2,000
Batch size	128
Training Epochs	500

I. Generalization to Unseen VRPs

To comprehensively evaluate URS’s zero-shot generalization, we test its performance on a diverse set of 99 unseen problems. This set is designed to cover both established and novel challenges, including 44 well-studied VRP variants and 55 less-explored VRP variants. The detailed results are presented in Table 15 and Table 16. On widely-studied constraints (i.e., C, O, L, TW, MD, B, BP), URS continues to deliver high-quality solutions. Notably, URS merely selects the next node from a given set of feasible candidates without any additional domain knowledge, making its performance especially meaningful. Benefiting from our UDR, URS can directly address 55 less-explored VRP variants that compound asymmetry and PD relations. Without any fine-tuning, URS still produces high-quality solutions for them, thereby strengthening its broad problem generality and deployment efficiency.

Table 15. Zero-shot generalization performance across 44 unseen well-studied VRP variants ($N = 100$).

Method	CVRPBP		OCVRPBP		CVRPBPL		OCVRPBPTW		CVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
MVMoE	13.95%	14s	18.27%	14s	13.38%	14s	9.42%	15s	16.66%	16s
MTPOMO	13.71%	7s	17.90%	8s	13.44%	8s	9.94%	9s	17.15%	10s
ReLD-MTL	13.57%	9s	15.81%	9s	12.96%	10s	8.43%	11s	15.95%	12s
URS	12.95%	7s	16.96%	7s	12.65%	8s	10.05%	8s	18.33%	10s
Method	CVRPBPTW		OCVRPBPL		OCVRPBPLTW		CVRPL		CVRPBL	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	16m	0.00%	3.5h
MVMoE	16.86%	15s	17.75%	14s	8.87%	16s	0.26%	12s	1.35%	12s
MTPOMO	17.22%	9s	17.31%	8s	9.47%	10s	0.48%	6s	1.79%	6s
ReLD-MTL	16.09%	11s	15.21%	10s	8.06%	11s	0.02%	8s	1.01%	7s
URS	18.15%	9s	16.37%	8s	9.51%	9s	0.43%	7s	1.65%	6s
Method	OCVRPB		CVRPLTW		OCVRPBPTW		CVRPBPTW		OCVRPL	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	3.5h	0.00%	3.5h	0.00%	3.5h	0.00%	3.5h	0.00%	3.5h
MVMoE	7.09%	12s	1.47%	16s	9.95%	15s	7.33%	16s	3.15%	13s
MTPOMO	7.34%	5s	1.92%	9s	10.45%	6s	7.75%	8s	3.44%	6s
ReLD-MTL	5.36%	6s	1.17%	11s	9.29%	8s	6.94%	9s	2.31%	8s
URS	9.35%	6s	2.67%	9s	13.77%	7s	9.18%	8s	3.22%	7s
Method	CVRPBPTW		OCVRPBL		OCVRPLTW		OCVRPBPTW		MDCVRPBP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	3.5h	0.00%	3.5h	0.00%	3.5h	0.00%	3.5h	0.00%	6.6m
MVMoE	7.08%	15s	7.12%	12s	3.90%	15s	10.01%	15s	65.98%	33s
MTPOMO	7.41%	7s	7.34%	7s	4.37%	8s	10.50%	7s	45.77%	24s
ReLD-MTL	6.74%	8s	5.41%	7s	3.16%	9s	9.22%	8s	40.70%	27s
URS	8.94%	8s	9.47%	7s	5.12%	9s	13.72%	8s	35.44%	24s
Method	MDOCVRPBP		MDCVRPBPL		MDOCVRPBPTW		MDCVRPBPTW		MDCVRPBPTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
MVMoE	56.60%	33s	65.81%	36s	63.77%	38s	68.49%	42s	69.00%	39s
MTPOMO	42.94%	24s	45.98%	26s	47.09%	29s	50.45%	32s	50.74%	30s
ReLD-MTL	37.74%	27s	39.57%	29s	45.16%	34s	45.23%	37s	45.62%	35s
URS	24.44%	22s	35.07%	26s	26.31%	26s	36.49%	32s	36.71%	30s
Method	MDOCVRPBPL		MDOCVRPBPTW		MDCVRP		MDCVRPTW		MDOCVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
MVMoE	56.26%	34s	62.47%	40s	33.06%	31s	53.64%	41s	29.08%	31s
MTPOMO	41.98%	27s	46.26%	32s	23.38%	23s	36.10%	31s	27.46%	23s
ReLD-MTL	37.45%	29s	44.74%	36s	18.39%	25s	30.90%	34s	22.32%	25s
URS	24.22%	24s	26.20%	28s	15.05%	23s	37.26%	32s	22.01%	23s
Method	MDCVRPL		MDCVRPB		MDOCVRPTW		MDOCVRPB		MDCVRPBL	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
MVMoE	33.42%	34s	21.70%	26s	51.57%	40s	28.21%	26s	20.35%	29s
MTPOMO	24.32%	25s	12.89%	19s	35.71%	29s	25.06%	20s	11.10%	20s
ReLD-MTL	18.48%	28s	9.38%	21s	33.55%	33s	18.50%	21s	8.26%	23s
URS	15.32%	26s	13.14%	19s	26.05%	25s	15.17%	18s	11.19%	21s
Method	MDCVRPLTW		MDOCVRPBPTW		MDCVRPBPTW		MDOCVRPL		MDCVRPBPTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
MVMoE	55.55%	44s	62.47%	35s	63.18%	37s	29.33%	33s	65.90%	35s
MTPOMO	37.34%	32s	45.55%	25s	45.07%	27s	27.21%	25s	45.94%	25s
ReLD-MTL	31.56%	36s	43.36%	28s	40.57%	30s	22.05%	27s	41.28%	28s
URS	38.04%	35s	30.32%	22s	37.70%	27s	22.15%	25s	37.78%	25s
Method	MDOCVRPBL		MDOCVRPLTW		MDOCVRPBPTW		SPCTSP			
	Gap	Time	Gap	Time	Gap	Time	Gap	Time		
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	1.6h	—	
MVMoE	28.70%	28s	51.09%	43s	62.10%	37s	—	—	—	
MTPOMO	25.01%	21s	35.39%	31s	45.48%	27s	—	—	—	
ReLD-MTL	18.50%	22s	33.26%	35s	43.27%	30s	—	—	—	
URS	14.90%	20s	25.59%	27s	30.41%	24s	-2.37%	5s	—	

Table 16. Zero-shot generalization performance across 55 unseen less-explored VRP variants ($N = 100$).

Method	ACVRPL		ACVRPBL		ACVRPLTW		ACVRPBLTW		AOCVRPL	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	-2.24%	1.9m	6.18%	2.4m	12.22%	2.7m	10.18%	2.4m	37.36%	3.1m
Method	AOCVRPBL		AOCVRPLTW		AOCVRPBLTW		ACVRPBPL		ACVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	17.23%	2.7m	16.35%	2.8m	14.32%	2.5m	24.03%	2.6m	13.82%	2.8m
Method	AOCVRPBPL		AOCVRPBPLTW		ACVRPB		ACVRPTW		AOCVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	32.50%	2.7m	16.43%	2.7m	7.10%	2.1m	11.28%	2.5m	37.63%	2.9m
Method	AOCVRPTW		AOCVRPB		AOCVRPBPTW		ACVRPBPTW		ACVRBPB	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	17.47%	2.6m	18.05%	2.5m	14.45%	2.3m	9.95%	2.2m	24.20%	2.4m
Method	AOCVRBPB		AOCVRPBPTW		ACVRPBPTW		APDTSP		AMDCVRPL	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	32.58%	2.5m	17.45%	2.5m	14.52%	2.6m	6.21%	1m	7.15%	6.1m
Method	AMDCVRPBL		AMDCVRPLTW		AMDCVRPBLTW		AMDOCVRPL		AMDOCVRPBL	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	17.51%	6.5m	19.98%	8.8m	16.73%	7.5m	47.80%	8.6m	27.23%	7.1m
Method	AMDOCVRPLTW		AMDOCVRPBLTW		AMDCVRPBPL		AMDCVRPBPLTW		AMDOCVRBPB	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	24.90%	7.9m	22.84%	6.7m	35.00%	7.2m	20.50%	8.3m	38.78%	7.3m
Method	AMDOCVRPBPLTW		AMDCVRP		AMDCVRPTW		AMDOCVRP		AMDCVRPB	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	24.05%	7.5m	8.11%	5.3m	20.96%	8.1m	48.02%	7.8m	17.35%	5.8m
Method	AMDOCVRPTW		AMDOCVRPB		AMDOCVRPBPTW		AMDCVRPBPTW		AMDCVRBPB	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	26.02%	7.3m	27.78%	6.5m	23.67%	6.2m	17.52%	6.8m	36.18%	6.5m
Method	AMDOCVRBPB		AMDOCVRPBPTW		AMDCVRPBPTW		PDCVRP		OPDCVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m
URS	39.31%	6.7m	24.99%	6.9m	20.98%	7.6m	-1.47%	4.3s	4.93%	4.3s
Method	APDCVRP		AOPDCVRP		AOP		APCTSP		ASPCTSP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Value	Time
Oracle	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	0.00%	6.6m	—	
URS	7.03%	1.2m	11.13%	1.2m	-5.47%	1.2m	43.79%	1.2m	2.22	1.2m

J. More Experiments

J.1. Training Problem Selection

To empirically validate our selection strategy of training problems, we train two additional variants and compare them with our original URS: URS-Diff (Different Selection) and URS-Expand (Broader Problems). The first variant, URS-Diff, uses a different problem set where OCVRPTW is replaced by attribute-independent CVRPL. This modification means that the Open Route and Time Window attributes each appear in only one training problem. The second variant, URS-Expand, is trained on a broader set of 22 problems, incorporating all variants evaluated in MTPOMO (Liu et al., 2024) and MVMoE (Zhou et al., 2024).

As shown in Table 17, URS-Diff performs significantly worse on unseen problems involving Open Route and Time Window constraints. Meanwhile, URS-Expand achieves a slightly better performance than URS due to seeing more problem variants during training. However, the original URS remains highly competitive. Remarkably, despite being trained on only half the number of problem types (11 vs. 22), URS achieves the best solution on 12 out of the 22 evaluated problems.

These results demonstrate that the choice of our selected 11 problems offers a balanced trade-off, providing sufficient attribute coverage for strong generalization while maintaining training efficiency. Expanding the training set yields only marginal gains, while a poorly selected set leads to poor cross-problem generalization. In future work, we plan to explore more effective training problem selection strategies to further enhance model performance and training efficiency.

Table 17. Comparison under different training problem settings. The symbol * indicates the original problem selection, and Δ represents a different problem selection of equal number. All test problems are encompassed within URS-Expand.

Method	ATSP(* Δ)	TSP(* Δ)	OP(* Δ)	PCTSP(* Δ)	PDTSP(* Δ)	ACVRP(* Δ)
URS-Diff	4.52%	1.05%	0.82%	1.31%	6.09%	5.10%
URS-Expand	4.77%	0.93%	0.99%	1.41%	5.85%	5.48%
URS	2.26%	0.57%	0.45%	1.06%	4.98%	3.06%
	CVRP(* Δ)	CVRPTW(* Δ)	CVRPB(* Δ)	OCVRP(* Δ)	OCVRPTW(*)	CVRPL(Δ)
URS-Diff	1.97%	6.26%	1.80%	3.75%	15.45%	0.58%
URS-Expand	2.04%	5.62%	1.77%	3.45%	4.34%	0.65%
URS	1.81%	6.13%	1.46%	3.24%	5.07%	0.43%
	OCVRPB	CVRPBL	CVRPLTW	OCVRPBTW	CVRPBLTW	OCVRPL
URS-Diff	7.55%	2.02%	2.76%	27.48%	10.25%	3.73%
URS-Expand	5.39%	1.88%	2.03%	9.13%	6.29%	3.41%
URS	9.35%	1.65%	2.67%	13.77%	9.18%	3.22%
	CVRPBTW	OCVRPBL	OCVRPLTW	OCVRPBLTW	Avg.gap	Best Sol.
URS-Diff	10.00%	7.56%	15.61%	27.80%	7.43%	0/22
URS-Expand	6.07%	5.47%	4.38%	9.23%	4.12%	10/22
URS	8.94%	9.47%	5.12%	13.72%	4.89%	12/22

J.2. Comparison with RouteFinder and CaDA

We strictly adhered to the problem settings established by MVMoE (Zhou et al., 2024) and ReLD (Huang et al., 2025) to ensure a fair and consistent evaluation. Notably, as shown in Table 6, URS already outperforms RF (Berto et al., 2025) and CaDA (Li et al., 2025a) on the large-scale CVRPLib benchmark dataset. In our cross-problem test, RF and CaDA are excluded due to discrepancies in the problems, including the number of trained CVRP variants (16 in their work vs. 5 in ours) and constraint settings (i.e., Backhauls and Time Windows). To ensure a fair comparison with RF and CaDA, we conduct two additional experiments:

- We evaluate URS against CaDA and the best-performing RouteFinder variant (RF-TE) on a shared subset of both seen and unseen problems, strictly restricting the evaluation to variants with perfectly aligned constraint definitions across all methods. As shown in Table 18, when evaluated under strictly matched conditions, URS still performs best overall among comparable methods. Note that URS can provide the strongest zero-shot generalization across all shared unseen problems.
- We retrain both RouteFinder-TE (denoted as RF-Retrain) and CaDA (denoted as CaDA-Retrain) from scratch using the exact same training settings as our URS and evaluate the three models on 48 CVRP variants. Since RF fails to train beyond 150 epochs due to the occurrence of NaN probabilities in our training setting, we evaluate all three methods at exactly 150 epochs to ensure a fair comparison. The experimental results in Table 19 show that under consistent training settings, URS still achieves the best overall performance among the three methods and provides the best solutions for the majority of problems (32/48), highlighting our strong zero-shot generalization.

Table 18. Comparison of shared seen and unseen VRP variants between URS, RF-TE, and CaDA.

Method	Seen		Unseen				Avg.gap	Best Sol.
	CVRP	OCVRP	MDCVRP	MDOCVRP	MDCVRPL	MDOCVRPL		
RF-TE	2.06%	2.86%	63.60%	33.52%	66.75%	34.40%	33.86%	0/6
CaDA	2.17%	2.80%	40.96%	53.21%	42.40%	53.34%	32.48%	1/6
URS	1.81%	3.24%	15.04%	22.01%	15.32%	22.15%	13.26%	5/6

J.3. Detailed Results on CVRPLib Set-X Dataset

We further evaluate the generalization ability of URS on benchmark instances from CVRPLib Set-X (Uchoa et al., 2017) with scale $\in [500, 1000]$. Our test results on CVRPLib Set-X datasets are presented in Table 20. All models are trained on instances of size $N = 100$. In the Set-X dataset, some results for comparative models are sourced from MVMoE (Zhou et al., 2024) and RouteFinder (Berto et al., 2025). As shown in Table 20, URS maintains its position as the best overall performance across instances of varying scales. Notably, RL-based URS surpasses the SL-based representative specialist model LEHD (8.678% vs. 12.836%) and many representative multi-task neural solvers (e.g., MVMoE and RouteFinder), underscoring its practical applicability in real-world scenarios.

Table 19. Comparison between URS and the retrained RouteFinder and CaDA on all 48 CVRP variants. Note that all models are trained for 150 epochs to ensure a fair comparison.

Method	CVRPTW		OCVRP		CVRP		CVRPB		OCVRPTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	8.38%	8s	6.03%	6s	3.58%	6s	4.47%	5s	8.18%	7s
CaDA-Retrain	5.28%	9s	3.22%	7s	1.92%	7s	1.61%	6s	4.42%	8s
URS	6.09%	8s	3.71%	6s	2.07%	6s	1.90%	5s	4.89%	8s

Method	OCVRPB		CVRPBL		CVRPLTW		OCVRPBWTW		CVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	11.61%	5s	4.73%	6s	4.77%	9s	14.17%	7s	10.11%	8s
CaDA-Retrain	8.46%	6s	4.42%	7s	4.63%	11s	10.39%	8s	10.87%	10s
URS	8.01%	6s	2.12%	6s	2.53%	9s	12.37%	7s	8.74%	8s

Method	OCVRPL		CVRPBWTW		OCVRPBL		OCVRPLTW		OCVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	5.98%	7s	9.74%	7s	11.58%	6s	8.34%	8s	14.46%	7s
CaDA-Retrain	5.81%	8s	7.17%	8s	12.48%	7s	8.55%	11s	15.94%	10s
URS	3.72%	7s	8.42%	7s	8.05%	6s	5.06%	8s	12.37%	8s

Method	CVRBPB		OCVRBPB		CVRPBPL		OCVRPBPTW		CVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	14.40%	7s	18.31%	7s	14.01%	7s	11.43%	9s	18.70%	10s
CaDA-Retrain	13.37%	8s	18.25%	8s	16.71%	9s	9.64%	10s	20.17%	12s
URS	13.37%	7s	16.08%	7s	12.94%	8s	10.07%	9s	18.47%	10s

Method	CVRPBPTW		OCVRPBPL		OCVRPBPLTW		CVRPL		MDCVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	18.84%	9s	17.76%	8s	10.88%	9s	2.21%	7s	27.10%	18s
CaDA-Retrain	17.11%	10s	21.44%	9s	13.54%	12s	2.29%	8s	27.09%	23s
URS	18.63%	9s	15.42%	8s	9.57%	10s	0.66%	7s	38.17%	23s

Method	MDCVRPTW		MDOCVRP		MDCVRPL		MDCVRPB		MDOCVRPTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	76.46%	25s	19.09%	18s	27.41%	34s	21.40%	15s	35.23%	22s
CaDA-Retrain	51.08%	30s	30.01%	19s	27.59%	21s	16.47%	19s	54.46%	29s
URS	38.25%	31s	21.34%	21s	39.22%	26s	20.83%	18s	28.40%	22s

Method	MDOCVRPB		MDCVRPBL		MDCVRPLTW		MDOCVRPBTW		MDCVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	17.84%	15s	20.02%	29s	79.33%	40s	44.11%	19s	82.19%	23s
CaDA-Retrain	26.72%	16s	15.02%	18s	43.43%	29s	64.29%	25s	50.11%	24s
URS	15.20%	17s	17.36%	21s	39.98%	33s	25.68%	19s	33.24%	27s

Method	MDOCVRPL		MDCVRPBWTW		MDOCVRPBL		MDOCVRPLTW		MDOCVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	19.02%	19s	84.94%	21s	17.72%	16s	35.06%	24s	44.12%	21s
CaDA-Retrain	30.61%	20s	61.39%	25s	26.94%	17s	39.49%	29s	49.12%	25s
URS	21.56%	24s	33.58%	25s	14.84%	19s	28.60%	24s	25.84%	21s

Method	MDCVRBPB		MDOCVRPBP		MDCVRPBPL		MDOCVRPBPTW		MDCVRPBPLTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
RF-Retrain	54.74%	21s	36.08%	21s	54.19%	23s	47.24%	26s	84.51%	30s
CaDA-Retrain	54.45%	25s	52.38%	22s	50.69%	24s	69.07%	32s	55.63%	32s
URS	41.49%	24s	24.18%	23s	41.08%	27s	24.19%	25s	33.99%	35s

Method	MDCVRPBPTW		MDOCVRPBPL		MDOCVRPBPLTW		Avg.gap		Best Solutions	
	Gap	Time	Gap	Time	Gap	Time				
RF-Retrain	85.12%	28s	36.06%	23s	47.63%	28s	29.24%		3/48	
CaDA-Retrain	66.78%	33s	53.48%	23s	53.06%	31s	28.45%		13/48	
URS	34.55%	32s	24.10%	25s	24.32%	27s	19.01%		32/48	

Table 20. Results on large-scale CVRPLib instances (Set-X) (Uchoa et al., 2017) ($500 \leq N \leq 1000$). All models are trained on instances of size $N = 100$.

Set-X		POMO		LEHD		MTPOMO		MVMoE/4E		MVMoE/4E-L		RF-MVMoE		RF-TE		URS	
Instance	Opt.	Obj.	Gap	Obj.	Gap	Obj.	Gap	Obj.	Gap	Obj.	Gap	Obj.	Gap	Obj.	Gap	Obj.	Gap
X-n502-k39	69226	75617	9.232%	71438	3.195%	77284	11.640%	73533	6.222%	74429	7.516%	76338	10.274%	71791	3.705%	71281	2.969%
X-n513-k21	24201	30518	26.102%	25624	5.880%	28510	17.805%	32102	32.647%	31231	29.048%	32639	34.866%	28465	17.619%	26166	8.119%
X-n524-k153	154593	201877	30.586%	280556	81.480%	192249	24.358%	186540	20.665%	182392	17.982%	170999	10.612%	174381	12.800%	175250	13.362%
X-n536-k96	94846	106073	11.837%	103785	9.425%	106514	12.302%	109581	15.536%	108543	14.441%	105847	11.599%	103272	8.884%	102969	8.564%
X-n548-k50	86700	103093	18.908%	90644	4.549%	94562	9.068%	95894	10.604%	95917	10.631%	104289	20.287%	100956	16.443%	89768	3.539%
X-n561-k42	42717	49370	15.575%	44728	4.708%	47846	12.007%	56008	31.114%	51810	21.287%	53383	24.969%	49454	15.771%	45964	7.601%
X-n573-k30	50673	83545	64.871%	53482	5.543%	60913	20.208%	59473	17.366%	57042	12.569%	61524	21.414%	55952	10.418%	54361	7.278%
X-n586-k159	190316	229887	20.792%	232867	22.358%	208893	9.761%	215668	13.321%	214577	12.748%	212151	11.473%	205575	8.018%	202645	6.478%
X-n599-k92	108451	150572	38.839%	115377	6.386%	120333	10.956%	128949	18.901%	125279	15.517%	126578	16.714%	116560	7.477%	114423	5.507%
X-n613-k62	59535	68451	14.976%	62484	4.953%	67984	14.192%	82586	38.718%	74945	25.884%	73456	23.383%	67267	12.987%	65901	10.693%
X-n627-k43	62164	84434	35.825%	67568	8.693%	73060	17.528%	70987	14.193%	70905	14.061%	70414	13.271%	67572	8.700%	66499	6.973%
X-n641-k35	63682	75573	18.672%	68249	7.172%	72643	14.071%	75329	18.289%	72655	14.090%	71975	13.023%	70831	11.226%	67005	5.218%
X-n655-k131	106780	127211	19.134%	117532	10.069%	116988	9.560%	117678	10.206%	118475	10.952%	119057	11.497%	112202	5.078%	110237	3.237%
X-n670-k130	146332	208079	42.197%	220927	50.977%	190118	29.922%	197695	35.100%	183447	25.364%	168226	14.962%	168999	15.490%	184010	25.748%
X-n685-k75	68205	79482	16.534%	72946	6.951%	80892	18.601%	97388	42.787%	89441	31.136%	82269	20.620%	77847	14.137%	75942	11.344%
X-n701-k44	81923	97843	19.433%	86327	5.376%	92075	12.392%	98469	20.197%	94924	15.870%	90189	10.090%	89932	9.776%	86038	5.023%
X-n716-k35	43373	51381	18.463%	46502	7.214%	52709	21.525%	56773	30.895%	52305	20.593%	52250	20.467%	49669	14.516%	46496	7.200%
X-n733-k159	136187	159098	16.823%	149115	9.493%	161961	18.925%	178322	30.939%	167477	22.976%	156387	14.833%	148463	9.014%	147743	8.485%
X-n749-k98	77269	87786	13.611%	83439	7.985%	90582	17.229%	100438	29.985%	94497	22.296%	92147	19.255%	85171	10.227%	83759	8.399%
X-n766-k71	114417	135464	18.395%	131487	14.919%	144041	25.891%	152352	33.155%	136255	19.086%	130505	14.061%	129935	13.563%	139371	21.810%
X-n783-k48	72386	90289	24.733%	76766	6.051%	83169	14.897%	100383	38.677%	92960	28.423%	96336	33.087%	83185	14.919%	77437	6.978%
X-n801-k40	73305	124278	69.536%	77546	5.785%	85077	16.059%	91560	24.903%	87662	19.585%	87118	18.843%	86164	17.542%	77369	5.544%
X-n819-k171	158121	193451	22.344%	178558	12.925%	177157	12.039%	183599	16.113%	185832	17.525%	179596	13.581%	174441	10.321%	171024	8.160%
X-n837-k142	193737	237884	22.787%	207709	7.212%	214207	10.566%	229526	18.473%	221286	14.220%	230362	18.904%	208528	7.635%	203457	5.017%
X-n856-k95	88965	152528	71.447%	92936	4.464%	101774	14.398%	99129	11.425%	106816	20.065%	105801	18.924%	98291	10.483%	94547	6.274%
X-n876-k59	99299	119764	20.609%	104183	4.918%	116617	17.440%	119619	20.463%	114333	15.140%	114016	14.821%	107416	8.174%	105417	6.161%
X-n895-k37	53860	70245	30.421%	58028	7.739%	65587	21.773%	79018	46.710%	64310	19.402%	69099	28.294%	64871	20.444%	58137	7.941%
X-n916-k207	329179	399372	21.324%	385208	17.021%	361719	9.885%	383681	16.557%	374016	13.621%	373600	13.494%	352998	7.236%	346556	5.279%
X-n936-k151	132715	237625	79.049%	196547	48.097%	186262	40.347%	220926	66.466%	190407	43.471%	161343	21.571%	163162	22.942%	172675	30.110%
X-n957-k87	85465	130850	53.104%	90295	5.651%	98198	14.898%	113882	33.250%	105629	23.593%	123633	44.659%	102689	20.153%	90485	5.874%
X-n979-k58	118976	147687	24.132%	127972	7.561%	138092	16.067%	146347	23.005%	139682	17.404%	131754	10.740%	129952	9.225%	125353	5.360%
X-n1001-k43	72355	100399	38.759%	76689	5.990%	87660	21.153%	114448	58.176%	94734	30.929%	88969	22.962%	85929	18.760%	77739	7.441%
Avg. Gap		29.658%		12.836%		16.796%		26.408%		19.607%		18.795%		12.303%		8.678%	

K. Ablation Study

In this section, we conduct a detailed ablation study and analysis to demonstrate the effectiveness and robustness of URS. Please note that, unless stated otherwise, the results presented in the ablation study reflect the best result from multiple trajectories, and we employ instance augmentation to improve performance in the ablation study. We adopt the widely used 100-node setting as our primary evaluation scenario for URS.

K.1. Effects of Node-type Indicator

Table 21. The ablation study of the node-type indicator on trained VRPs

Method	ATSP	TSP	OP	PCTSP	PDTSP	ACVRP	CVRP	CVRPTW	CVRPB	OCVRP	OCVRPTW	Avg.gap
w/o ξ_i	2.14%	0.48%	0.45%	1.06%	9.28%	2.84%	3.66%	6.61%	1.57%	7.87%	5.66%	3.78%
w/ ξ_i	2.26%	0.57%	0.45%	1.06%	4.98%	3.06%	1.81%	6.13%	1.46%	3.24%	5.07%	2.74%

To evaluate the effectiveness of node-type indicator ξ , we conduct an ablation study about including and excluding ξ . Table 21 shows that adding the node-type indicator reduces the average optimality gap from 3.78% to 2.74%, with large gains on the routing problem with relatively rich instance node roles, such as PDTSP, where the coexistence of a depot and paired pickup-delivery nodes (with inherent precedence coupling) allows the explicit node-role encoding to further boost performance. Meanwhile, URS maintains a competitive performance in addressing the problem of single-node roles.

K.2. Effects of Problem State

Table 22. Performance impact of removing the problem state feature C_t

Method	ATSP	TSP	OP	PCTSP	PDTSP	ACVRP	CVRP	CVRPTW	CVRPB	OCVRP	OCVRPTW	Avg.gap
w/o C_t	2.09%	0.45%	0.60%	1.20%	4.59%	4.03%	2.70%	5.71%	2.64%	4.02%	4.62%	2.97%
w/ C_t	2.26%	0.57%	0.45%	1.06%	4.98%	3.06%	1.81%	6.13%	1.46%	3.24%	5.07%	2.74%

As described in Appendix E.3, the majority of VRP variants are already solved without supplying an additional problem state C_t . To test whether the remaining use of C_t matters, we conduct an ablation in which we eliminate it entirely, enforcing $C_t \equiv 0$ for all problems. The results in Table 22 show that URS maintains its competitive performance even in the absence of C_t . This indicates that explicit constraint-state features are not essential to URS effectiveness. Interestingly, on (O)CVRPTW instances, the model without any explicit constraint features outperforms the version that supplies only capacity information. We guess that capacity and time windows information are equally important, so enforcing only capacity creates a bias toward load considerations.

K.3. Effects of Mixed Bias Module

Table 23. Ablation of prior components in the MBM.

D	D^T	R	ATSP	TSP	OP	PCTSP	PDTSP	ACVRP	CVRP	CVRPTW	CVRPB	OCVRP	OCVRPTW	Avg.gap
✓	×	×	16.64%	0.44%	0.25%	0.67%	7.33%	16.62%	2.42%	6.44%	3.96%	9.44%	5.47%	6.33%
✓	✓	×	2.09%	0.45%	0.48%	0.82%	7.66%	9.11%	2.52%	6.89%	2.68%	5.70%	6.36%	4.07%
✓	×	✓	16.60%	0.46%	0.49%	0.94%	4.49%	26.22%	4.16%	6.57%	3.07%	12.23%	5.53%	7.34%
✓	✓	✓	2.26%	0.57%	0.45%	1.06%	4.98%	3.06%	1.81%	6.13%	1.46%	3.24%	5.07%	2.74%

To further assess the effectiveness of MBM, we conduct an ablation study that examines different combinations of problem-specific priors within MBM. Owing to its flexibility, the MBM allows multiple priors to be imposed on the same shared attention layer. The results in Table 23 underscore the complementary roles of the three prior components: (1) The transposed distance matrix D^T captures directional asymmetry absent from the raw distance matrix, and its inclusion markedly stabilizes optimization on asymmetric variants; (2) Removing the pickup-delivery relation matrix R degrades performance on PDTSPs, confirming that explicit relational coupling is indispensable for these instances; (3) When all priors are fused, MBM can simultaneously learn the geometric and relational biases inherent in various problems, yielding the best overall performance.

K.4. Effects of Different Decoder Architectures

Table 24. Effect of different decoder architectures on URS cross-problem performance.

Decoder	ATSP	TSP	OP	PCTSP	PDTSP	ACVRP	CVRP	CVRPTW	CVRPB	OCVRP	OCVRPTW	Avg.gap
URS-POMO	4.83%	0.74%	0.72%	1.29%	5.45%	5.09%	1.99%	6.42%	1.83%	3.62%	5.31%	3.39%
URS-ReLD	6.48%	0.90%	0.86%	1.51%	5.81%	6.51%	2.33%	6.56%	2.33%	3.62%	5.64%	3.87%
URS	2.26%	0.57%	0.45%	1.06%	4.98%	3.06%	1.81%	6.13%	1.46%	3.24%	5.07%	2.74%

We adopt an AM (Kool et al., 2019) as our basic encoder-decoder model. Multiple decoder refinements have been proposed around AM (Kwon et al., 2020; Huang et al., 2025; Zhou et al., 2026a). To assess how decoder structure impacts multi-task performance, we additionally train two URS variants whose decoders follow POMO and ReLD, denoted URS-POMO and URS-ReLD. For stronger and fairer baselines, we apply Distance-aware Attention Reshaping (DAR) (Wang et al., 2025b) to both, and all decoder parameters are generated by a hypernetwork conditioned on the problem representation λ (see Appendix E.2). As shown in Table 24, URS-POMO and URS-ReLD still exhibit strong cross-problem robustness, while replacing the decoder with ICAM (Zhou et al., 2026a) yields a further overall improvement.

Furthermore, the results in Table 24 reveal a notable optimization challenge. By incorporating an additional Feed-Forward Network layer compared to POMO, the ReLD decoder forces the hypernetwork to map to a larger parameter space, thereby increasing its learning difficulty and degrading performance. This optimization difficulty prevents URS-ReLD from matching the performance of the original ReLD-MTL, where weights are learned directly rather than being dynamically generated. Given the outstanding performance of ReLD and other heavy-decoder paradigms in single-task optimization (Luo et al., 2023; Drakulic et al., 2023), exploring how to adapt problem-conditioned generation mechanisms for heavier decoders is a highly promising direction.

K.5. Effects of MBM & Parameter Generator on Zero-shot Generalization

 Table 25. Ablation of MBM, WEIGHT(λ), and BIAS(λ) on cross-problem zero-shot generalization. The symbol (*) indicates the seen problems in training.

MBM	WEIGHT(λ)	BIAS(λ)	ATSP (*)	TSP (*)	OP (*)	PCTSP (*)	PDTSP (*)	ACVRP (*)
×	✓	✓	16.64%	0.44%	0.25%	0.67%	7.33%	16.62%
✓	×	×	4.24%	1.10%	1.04%	1.42%	6.38%	4.07%
✓	×	✓	3.83%	1.16%	1.09%	1.52%	6.65%	4.27%
✓	✓	×	2.79%	1.58%	1.39%	1.92%	7.05%	3.39%
✓	✓	✓	2.26%	0.57%	0.45%	1.06%	4.98%	3.06%

MBM	WEIGHT(λ)	BIAS(λ)	CVRP (*)	CVRPTW (*)	CVRPB (*)	OCVRP (*)	OCVRPTW (*)	CVRPL
×	✓	✓	2.42%	6.44%	3.96%	9.44%	5.47%	0.98%
✓	×	×	6.23%	7.22%	5.00%	3.99%	5.41%	2.92%
✓	×	✓	4.94%	6.08%	3.00%	4.61%	5.21%	4.79%
✓	✓	×	2.90%	6.60%	3.12%	4.47%	4.76%	1.46%
✓	✓	✓	1.81%	6.13%	1.46%	3.24%	5.07%	0.43%

MBM	WEIGHT(λ)	BIAS(λ)	OCVRPB	CVRPBL	CVRPLTW	OCVRPBTW	CVRPBLTW	OCVRPL
×	✓	✓	15.95%	4.18%	2.89%	18.78%	10.93%	9.51%
✓	×	×	9.54%	5.30%	2.73%	14.37%	9.80%	4.64%
✓	×	✓	9.58%	3.25%	3.07%	14.16%	9.91%	3.88%
✓	✓	×	9.92%	3.34%	3.08%	14.02%	9.92%	4.52%
✓	✓	✓	9.35%	1.65%	2.67%	13.77%	9.18%	3.22%

MBM	WEIGHT(λ)	BIAS(λ)	CVRPBTW	OCVRPBL	OCVRPLTW	OCVRPBLTW	Avg.gap	Best Sol.
×	✓	✓	10.74%	16.14%	5.57%	18.80%	8.37%	3/22
✓	×	×	9.67%	10.67%	5.23%	14.49%	6.16%	0/22
✓	×	✓	9.63%	10.42%	5.51%	13.99%	5.93%	1/22
✓	✓	×	9.79%	10.02%	5.81%	13.98%	5.72%	1/22
✓	✓	✓	8.94%	9.47%	5.12%	13.72%	4.89%	17/22

To empirically evaluate the necessity of these components, we have conducted tests on the 11 seen problems as well as 11

additional unseen CVRP variants widely investigated in current multi-task research (e.g., MVMoE (Zhou et al., 2024)). The results are presented in Table 25. We can observe that removing MBM leads to a significant performance drop on problems with complex geometric properties, such as ATSP, ACVRP, and PDTSP, which confirms MBM is critical for perceiving specific geometric biases. In addition, removing WEIGHT(λ) or BIAS(λ) also results in performance degradation across most tasks. In terms of overall performance, URS significantly outperforms any model where a single module is removed. The full URS model achieves the lowest average gap and provides the best solution on the majority of tasks (17/22). This demonstrates that the synergy of these components is essential for robust cross-problem generalization.

K.6. Effects of Each UDR Component

To clarify the independent contribution of each UDR component, we conduct ablations by removing each component in turn. As shown in Table 26, the absence of any single component leads to notable performance degradation. A complete UDR achieves the lowest average gap and provides the best solution on the majority of tasks (15/22), affirming the necessity of our UDR design for robust zero-shot generalization.

Table 26. The ablation study of each UDR component. The symbol (*) indicates the seen problems in training. ρ , ω , and ξ represent positional identifier, unified attribute set, and node-type indicator, respectively. Since removing positional identifiers prevents the use of instance augmentation, we have excluded it from all results to ensure consistency in the inference strategy.

ρ	ω	ξ	ATSP (*)	TSP (*)	OP (*)	PCTSP (*)	PDTSP (*)	ACVRP (*)
×	✓	✓	7.64%	1.76%	1.17%	1.96%	8.54%	7.16%
✓	×	✓	10.07%	1.98%	2.70%	7.05%	13.50%	13.36%
✓	✓	×	5.43%	1.02%	1.08%	1.86%	12.72%	5.81%
✓	✓	✓	5.51%	1.17%	0.96%	1.79%	7.71%	6.11%
ρ	ω	ξ	CVRP (*)	CVRPTW (*)	CVRPB (*)	OCVRP (*)	OCVRPTW (*)	CVRPL
×	✓	✓	2.68%	6.88%	2.84%	4.64%	5.91%	1.29%
✓	×	✓	4.15%	23.71%	5.24%	7.10%	19.10%	2.70%
✓	✓	×	4.72%	7.67%	2.83%	9.64%	6.96%	3.33%
✓	✓	✓	2.62%	6.80%	2.69%	4.54%	6.00%	1.23%
ρ	ω	ξ	OCVRPB	CVRPBL	CVRPLTW	OCVRPBTW	CVRPBLTW	OCVRPL
×	✓	✓	11.87%	3.12%	3.34%	34.34%	44.03%	4.63%
✓	×	✓	15.02%	5.47%	19.53%	30.88%	29.64%	6.98%
✓	✓	×	26.06%	2.69%	4.13%	34.62%	26.55%	9.68%
✓	✓	✓	11.54%	3.03%	3.38%	15.47%	10.70%	4.55%
ρ	ω	ξ	CVRPBTW	OCVRPBL	OCVRPLTW	OCVRPBLTW	Avg.gap	Best Sol.
×	✓	✓	43.99%	11.99%	6.02%	34.66%	11.38%	3/22
✓	×	✓	29.34%	14.84%	19.22%	31.02%	14.21%	0/22
✓	✓	×	28.60%	25.96%	7.00%	35.93%	12.01%	4/22
✓	✓	✓	10.51%	11.68%	6.08%	15.38%	6.34%	15/22

K.7. Hypernetwork vs. Adapter

We train URS-Adapter, a URS variant that adopts a distinct decoder for each of the 11 variants, and compare it with the original URS using a hypernetwork on 11 seen problems (see Table 27) and large-scale CVRP variants (see Table 28). URS consistently outperforms URS-Adapter across 11 seen problems, achieving higher parameter efficiency. This performance advantage holds consistently on large-scale CVRP variants, which further solidifies our conclusion.

L. Licenses for Used Resources

We list the existing codes and datasets in Table 29, all of which are open-source resources for academic use.

Table 27. Performance comparison between URS-Hypernet and URS-Adapter on 11 seen VRP variants.

Method	OP		PCTSP		PDTSP		ACVRP	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time
URS-Adapter	0.88%	4s	1.47%	4s	5.69%	4s	3.83%	1.5m
URS	0.45%	4s	1.06%	4s	4.98%	4s	3.06%	1.5m

Method	CVRPTW		CVRPB		OCVRP		OCVRPTW	
	Gap	Time	Gap	Time	Gap	Time	Gap	Time
URS-Adapter	6.34%	8s	3.21%	5s	4.06%	6s	4.94%	7s
URS	6.13%	8s	1.46%	5s	3.24%	6s	5.07%	7s

Method	ATSP		TSP		CVRP		Average	Best
	Gap	Time	Gap	Time	Gap	Time	Gap	Solution
URS-Adapter	2.97%	1.2m	0.82%	4s	2.85%	6s	3.37%	1/11
URS	2.26%	1.2m	0.57%	4s	1.81%	6s	2.74%	10/11

Table 28. Generalization comparison between URS and URS-Adapter across diverse large-scale CVRP variants.

Problem	Method	N = 1000		N = 2000		N = 3000		N = 4000		N = 5000	
		Obj.(Gap)	Time	Obj.(Gap)	Time	Obj.(Gap)	Time	Obj.(Gap)	Time	Obj.(Gap)	Time
CVRPTW	PyVRP	159.35(0.00%)	20m	337.15(0.00%)	40m	516.97(0.00%)	1h	618.58(0.00%)	2.7h	787.25(0.00%)	3.3h
	URS-Adapter	173.68(8.99%)	1.3m	367.22(8.92%)	10.8m	564.55(9.21%)	30.8m	677.91(9.59%)	1.2h	860.63(9.32%)	2.3h
	URS	172.58(8.30%)	1.3m	365.27(8.34%)	11.1m	559.03(8.14%)	30.1m	673.63(8.90%)	1.2h	853.79(8.45%)	2.3h
CVRPB	PyVRP	36.52(0.00%)	20m	51.80(0.00%)	40m	75.12(0.00%)	1h	93.52(0.00%)	2.7h	112.21(0.00%)	3.3h
	URS-Adapter	42.71(16.96%)	43s	62.73(21.11%)	5.5m	85.72(14.11%)	17.5m	107.17(14.59%)	41.9m	127.93(14.01%)	1.3h
	URS	34.86(-4.54%)	40s	50.39(-2.72%)	5.1m	72.01(-4.15%)	17.3m	91.74(-1.91%)	41.2m	111.93(-0.24%)	1.3h
OCVRPTW	PyVRP	90.91(0.00%)	20m	164.00(0.00%)	40m	224.16(0.00%)	1h	299.45(0.00%)	2.7h	367.86(0.00%)	3.3h
	URS-Adapter	105.96(16.56%)	1.1m	203.52(24.10%)	9.9m	287.70(28.35%)	28.6m	391.06(30.59%)	1h	483.19(31.35%)	1.9h
	URS	98.76(8.63%)	1m	178.07(8.58%)	7.6m	243.88(8.80%)	25.4m	325.64(8.74%)	1h	398.72(8.39%)	1.9h

Table 29. List of licenses for the codes and datasets we used in this work.

Resource	Type	Link	License
PyVRP (Wouda et al., 2024)	Code	https://github.com/PyVRP/PyVRP	MIT License
LKH3 (Helsgaun, 2017)	Code	http://webhotel4.ruc.dk/~keld/research/LKH-3/	Available for academic research use
OR-Tools (Perron & Fumon, 2023)	Code	https://github.com/google/or-tools	Apache-2.0 License
POMO (Kwon et al., 2020)	Code	https://github.com/yd-kwon/POMO/tree/master/NEW_py_ver	MIT License
Sym-NCO (Kim et al., 2022)	Code	https://github.com/alstn12088/Sym-NCO	Available for any non-commercial use
LEHD (Luo et al., 2023)	Code	https://github.com/CIAM-Group/NCO_code/tree/main/single_objective/LEHD	Available for any non-commercial use
BQ (Drakulic et al., 2023)	Code	https://github.com/naver/bq-nco	CC BY-NC-SA 4.0 license
ICAM (Zhou et al., 2026a)	Code	https://github.com/CIAM-Group/ICAM	MIT License
ELG (Gao et al., 2024)	Code	https://github.com/gaocr/ELG	MIT License
MatNet (Kwon et al., 2021)	Code	https://github.com/yd-kwon/MatNet/tree/main	MIT License
Heter-AM (Li et al., 2021)	Code	https://github.com/jingwenli0312/Heterogeneous-Attentions-PDP-DRL	MIT License
MTPOMO (Liu et al., 2024)	Code	https://github.com/Feiliu36/MTNCO	MIT License
MVMoE (Zhou et al., 2024)	Code	https://github.com/RoyalSkye/Routing-MVMoE	MIT License
RouteFinder (Berto et al., 2025)	Code	https://github.com/ai4co/routefinder	MIT License
CaDA (Li et al., 2025a)	Code	https://github.com/CIAM-Group/CaDA	MIT License
ReLD-MTL (Huang et al., 2025)	Code	https://github.com/ziweileonhuang/reld-nco	MIT License
GOAL (Drakulic et al., 2025)	Code	https://github.com/naver/goal-co	Available for any non-commercial use
CVRPLib Set-X (Uchoa et al., 2017)	Dataset	http://vrp.galgos.inf.puc-rio.br/index.php/en/	Available for academic research use
CVRPLib Set-AGS (Arnold et al., 2019)	Dataset	http://vrp.galgos.inf.puc-rio.br/index.php/en/	Available for academic research use